Prove The following statements :

निम्न को सिद्ध कीजिये :

1. $\cos^4 A - \sin^4 A + 1 = 2\cos^2 A$.

Ans.

Sol. L.H.S. $= \cos^4 A - \sin^4 A + 1$

$$= \left(\cos^2 A - \sin^2 A\right)\left(\cos^2 A + \sin^2 A\right) + 1$$

$$= \left(\cos^2 A - \sin^2 A\right) + 1$$

$$= \cos^2 A - \left(1 - \cos^2 A\right) + 1$$

$$= \cos^2 A - 1 + \cos^2 A + 1$$

$$= \mathbf{2}\cos^2 A$$

2. $(\sin A + \cos A)(1 - \sin A \cos A) = \sin^3 A + \cos^3 A$

Ans.

Sol. R.H.S. $= \sin^3 A + \cos^3 A$

$$= (\sin A + \cos A)\left(\sin^2 A + \cos^2 A - \sin A \cos A\right)$$

$$= (\sin A + \cos A)(1 - \sin A \cos A)$$

3. $\dfrac{\sin A}{1+\cos A} + \dfrac{1+\cos A}{\sin A} = 2\,\text{cosec}\, A$

Ans.

Sol. L.H.S. $= \dfrac{\sin A}{1+\cos A} + \dfrac{1+\cos A}{\sin A} = \dfrac{\sin^2 A + (1+\cos A)^2}{\sin A(1+\cos A)}$

$$= \frac{\sin^2 A + 1 + 2\cos A + \cos^2 A}{(1+\cos A)\sin A} = \frac{2 + 2\cos A}{(1+\cos A)\sin A}$$

$$= \frac{2(1+\cos A)}{(1+\cos A)\sin A} = \frac{2}{\sin A}$$

$$= 2\,\text{cosec}\, A$$

4. $\sqrt{\frac{1-\sin A}{1+\sin A}} = \sec A - \tan A$

Ans.

Sol. L.H.S. $= \sqrt{\left(\frac{1-\sin A}{1+\sin A}\right)}$

$$= \sqrt{\left(\frac{1-\sin A}{1+\sin A}\right)} \times \sqrt{\left(\frac{1-\sin A}{1-\sin A}\right)}$$

$$= \frac{\sqrt{(1-\sin A)^2}}{\sqrt{\left(1-\sin^2 A\right)}} = \frac{1-\sin A}{\sqrt{\left(\cos^2 A\right)}} = \frac{1-\sin A}{\cos A}$$

$$= \frac{1}{\cos A} - \frac{\sin A}{\cos A}$$

$$= \sec A - \tan A$$

5. $\frac{\cos ec\, A}{\cos ec\, A - 1} + \frac{\cos ec\, A}{\cos ec\, A + 1} = 2\sec^2 A$

Ans.

Sol. L.H.S. $= \frac{\text{cosec}\, A}{\text{cosec}\, A - 1} + \frac{\text{cosec}\, A}{\text{cosec}\, A + 1}$

$$= \frac{\text{cosec}\, A(\text{cosec}\, A + 1) + (\text{cosec}\, A - 1)\,\text{cosec}\, A}{(\text{cosec}\, A - 1)(\text{cosec}\, A + 1)}$$

$$= \frac{\text{cosec}\, A(\text{cosec}\, A + 1 + \text{cosec}\, A - 1)}{\left(\text{cosec}^2 A - 1\right)}$$

$$= \frac{2\,\text{cosec}^2 A}{1 + \cot^2 A - 1} = \frac{2\,\text{cosec}^2 A}{\cot^2 A}$$

$$= \frac{2}{\sin^2 A} \times \frac{\sin^2 A}{\cos^2 A} = \frac{2}{\cos^2 A}$$

$$= 2\sec^2 A$$

6. $\dfrac{\cos ec\, A}{\cot A + \tan A} = \cos A$

Ans.

Sol. L.H.S. $= \dfrac{\text{cosec}\, A}{\cot A + \tan A}$

$$= \frac{\dfrac{1}{\sin A}}{\dfrac{\cos A}{\sin A} + \dfrac{\sin A}{\cos A}}$$

$$= \frac{\dfrac{1}{\sin A}}{\dfrac{\cos^2 A + \sin^2 A}{\sin A \cos A}}$$

$$= \cos A$$

7. $\dfrac{\sec A - \tan A}{\sec A + \tan A} = 1 - 2\sec A \tan A + 2\tan^2 A$

Ans.

Sol. L.H.S. $= \dfrac{\sec A - \tan A}{\sec A + \tan A}$

Multiply by $(\sec A - \tan A)$ both numerator and denominator, we get

L.H.S. $= \dfrac{(\sec A - \tan A)^2}{\sec^2 A - \tan^2 A}$

$$= \frac{\sec^2 A - 2\sec A \tan A + \tan^2 A}{1 + \tan^2 A - \tan^2 A}$$

$$= 1 + \tan^2 A - 2\sec A \tan A + \tan^2 A$$

$$= 1 - 2\sec A \tan A + 2\tan^2 A$$

8. $\dfrac{\tan A}{1 - \cot A} + \dfrac{\cot A}{1 - \tan A} = \sec A \cos ecA + 1$

Ans.

Sol. L.H.S. $= \dfrac{\dfrac{\sin A}{\cos A}}{1-\dfrac{\cos A}{\sin A}} + \dfrac{\dfrac{\cos A}{\sin A}}{1-\dfrac{\sin A}{\cos A}}$

$= \dfrac{\dfrac{\sin A}{\cos A}}{\dfrac{\sin A - \cos A}{\sin A}} + \dfrac{\dfrac{\cos A}{\sin A}}{\dfrac{\cos A - \sin A}{\cos A}}$

$= \dfrac{\sin^2 A}{(\sin A - \cos A)\cos A} + \dfrac{\cos^2 A}{(\cos A - \sin A)\sin A}$

$= \dfrac{\sin^3 A - \cos^3 A}{(\sin A - \cos A)\sin A \cos A}$

$= \dfrac{(\sin A - \cos A)(\sin^2 A + \cos^2 A + \sin A \cos A)}{\sin A \cos A(\sin A - \cos A)}$

$= \dfrac{\sin A \cos A + 1}{\sin A \cos A} = \dfrac{\sin A \cos A}{\sin A \cos A} + \dfrac{1}{\sin A \cos A}$

$= 1 + \sec A \operatorname{cosec} A$

9. $\dfrac{\cos A}{1-\tan A} + \dfrac{\sin A}{1-\cot A} = \sin A + \cos A$

Ans.

Sol. L.H.S. $= \dfrac{\cos A}{1-\tan A} + \dfrac{\sin A}{1-\cot A}$

$= \dfrac{\cos A}{1-\dfrac{\sin A}{\cos A}} + \dfrac{\sin A}{1-\dfrac{\cos A}{\sin A}}$

$= \dfrac{\cos A}{\dfrac{\cos A - \sin A}{\cos A}} + \dfrac{\sin A}{\dfrac{\sin A - \cos A}{\sin A}}$

$= \dfrac{\cos^2 A}{\cos A - \sin A} + \dfrac{\sin^2 A}{\sin A - \cos A}$

$= \dfrac{\cos^2 A - \sin^2 A}{\cos A - \sin A}$

$= \sin A + \cos A$

10. $(\sin A + \cos A)(\cot A + \tan A) = \sec A + \operatorname{cosec} A$

Ans.

Sol. L.H.S. $= (\sin A + \cos A)(\cot A + \tan A)$

$$= (\sin A + \cos A)\left(\frac{\cos A}{\sin A} + \frac{\sin A}{\cos A}\right)$$

$$= (\sin A + \cos A)\left(\frac{\cos^2 A + \sin^2 A}{\sin A \times \cos A}\right)$$

$$= \frac{\sin A + \cos A}{\sin A \cdot \cos A} = \frac{\sin A}{\sin A \cos A} + \frac{\cos A}{\sin A \cos A}$$

$$= \frac{1}{\cos A} + \frac{1}{\sin A} = \sec A + \operatorname{cosec} A$$

11. $\sec^4 A - \sec^2 A = \tan^4 A + \tan^2 A$

Ans.

Sol. L.H.S. $= \sec^4 A - \sec^2 A$

$$= \left(1 + \tan^2 A\right)^2 - \left(1 + \tan^2 A\right)$$

$$= \left(1 + 2\tan^2 A + \tan^4 A - 1 - \tan^2 A\right)$$

$$= \tan^4 A + \tan^2 A$$

12. $\sec^2 A \operatorname{cosec}^2 A = \tan^2 A + \cot^2 A + 2$

Ans.

Sol. L.H.S. $= \sec^2 A \operatorname{cosec}^2 A$

$$= \left(1 + \tan^2 A\right)\left(1 + \cot^2 A\right)$$

$$= 1 + \tan^2 A + \cot^2 A + \tan^2 A \cot^2 A$$

$$= 2 + \tan^2 A + \cot^2 A$$

13. $\tan^2 A - \sin^2 A = \sin^4 A \sec^2 A$

Ans.

Sol. L.H.S. $= \tan^2 A - \sin^2 A$

$$= \frac{\sin^2 A}{\cos^2 A} - \sin^2 A = \frac{\sin^2 A - \sin^2 A \cos^2 A}{\cos^2 A}$$

$$= \frac{\sin^2 A\left(1 - \cos^2 A\right)}{\cos^2 A} = \frac{\sin^2 A}{\cos^2 A} \cdot \sin^2 A$$

$$= \sin^4 A \cdot \frac{1}{\cos^2 A}$$

$$= \sin^4 A \cdot \sec^2 A$$

14. $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$

Ans.

Sol. L.H.S. $= (1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A)$

$$= \left(1 + \frac{\cos A}{\sin A} - \frac{1}{\sin A}\right)\left(1 + \frac{\sin A}{\cos A} + \frac{1}{\cos A}\right)$$

$$= \left(\frac{\sin A + \cos A - 1}{\sin A}\right)\left(\frac{\cos A + \sin A + 1}{\cos A}\right)$$

$$= \frac{\sin^2 A + \cos^2 A - 1 + 2\sin A \cos A}{\sin A \cos A}$$

$$= \frac{1 - 1 + 2\sin A \cos A}{\sin A \cos A} = \frac{2\sin A \cos A}{\sin A \cos A}$$

$$= 2$$

15. $\dfrac{1}{\operatorname{cosec} A - \cot A} - \dfrac{1}{\sin A} = \dfrac{1}{\sin A} - \dfrac{1}{\operatorname{cosec} A + \cot A}$

Ans.

Sol. R.H.S. $= \dfrac{1}{\sin A} - \dfrac{1}{\dfrac{1}{\sin A} + \dfrac{\cos A}{\sin A}} = \dfrac{1}{\sin A} - \dfrac{1}{\dfrac{1 + \cos A}{\sin A}}$

$$= \frac{1}{\sin A} - \frac{\sin A}{1 + \cos A} = \frac{1 + \cos A - \sin^2 A}{\sin A(1 + \cos A)}$$

$$= \frac{1 + \cos A - \left(1 - \cos^2 A\right)}{\sin A(1 + \cos A)} = \frac{\cos A + \cos^2 A}{\sin A(1 + \cos A)}$$

$$= \frac{\cos A(1+\cos A)}{\sin A(1+\cos A)} = \cot A$$

L.H.S. $= \dfrac{1}{\text{cosec } A - \cot A} - \dfrac{1}{\sin A}$

$$= \frac{\text{cosec } A + \cot A}{(\text{cosec } A - \cos A)(\text{cosec } A + \cot A)} - \text{cosec } A$$

$$= \frac{\text{cosec } A + \cot A}{\text{cosec}^2 A - \cot^2 A} - \text{cosec } A$$

$$= \frac{\text{cosec } A + \cot A}{1 + \cot^2 A - \cot^2 A} - \text{cosec } A$$

$= \text{cosec } A + \cot A - \text{cosec } A = \cot A$

L.H.S. = R.H.S.

16. $\dfrac{\cot A \cos A}{\cot A + \cos A} = \dfrac{\cot A - \cos A}{\cot A \cos A}$

Ans.

Sol. L.H.S. $= \dfrac{\cot A \cos A}{\cot A + \cos A} \cdot \left(\dfrac{\cot A - \cos A}{\cot A - \cos A} \right)$

$$= \frac{\cot A \cdot \cos A(\cot A - \cos A)}{\cot^2 A - \cos^2 A}$$

$$= \frac{\cot A \cdot \cos A(\cot A - \cos A)}{\dfrac{\cos^2 A}{\sin^2 A} - \cos^2 A}$$

$$= \frac{\cot A \cdot \cos A(\cot A - \cos A)}{\cos^2 A\left(\dfrac{1}{\sin^2 A} - 1 \right)}$$

$$= \frac{\cot A \cdot \cos A(\cot A - \cos A)}{\cos^2 A(\text{cosec}^2 A - 1)}$$

$$= \frac{\cot A \cdot \cos A(\cot A - \cos A)}{\cos^2 A \cdot \cot^2 A}$$

$$= \frac{\cot A - \cos A}{\cos A \cdot \cot A}$$

17. $\dfrac{\cot A + \tan B}{\cot B + \tan A} = \cot A \tan B$

Ans.

Sol. L.H.S. $= \dfrac{\cot A + \tan B}{\cot B + \tan A} = \dfrac{\cot A + \tan B}{\dfrac{1}{\tan B} + \dfrac{1}{\cot A}}$

$$= \frac{\cot A + \tan B}{\dfrac{\cot A + \tan B}{\cot A \cdot \tan B}}$$

$$= \cot A \cdot \tan B.$$

18. $\left(\dfrac{1}{\sec^2\alpha - \cos^2\alpha} + \dfrac{1}{\cos ec^2\alpha - \sin^2\alpha}\right)\cos^2\alpha \sin^2\alpha = \dfrac{1-\cos^2\alpha \sin^2\alpha}{2+\cos^2\alpha \sin^2\alpha}$

Ans.

Sol. L.H.S. $= \left(\dfrac{1}{\sec^2\alpha - \cos^2\alpha} + \dfrac{1}{\text{cosec}^2\alpha - \sin^2\alpha}\right)\cos^2\alpha \sin^2\alpha$

$$= \left[\frac{1}{\dfrac{1}{\cos^2\alpha} - \cos^2\alpha} + \frac{1}{\dfrac{1}{\sin^2\alpha} - \sin^2\alpha}\right]\cos^2\alpha \sin^2\alpha$$

$$= \left[\frac{1}{\dfrac{1-\cos^4\alpha}{\cos^2\alpha}} + \frac{1}{\dfrac{1-\sin^4\alpha}{\sin^2\alpha}}\right]\cos^2\alpha \sin^2\alpha$$

$$= \left[\frac{\cos^2\alpha}{1-\cos^4\alpha} + \frac{\sin^2\alpha}{1-\sin^4\alpha}\right]\cos^2\alpha \sin^2\alpha$$

$$= \left[\frac{\cos^2\alpha}{(1-\cos^2\alpha)(1+\cos^2\alpha)} + \frac{\sin^2\alpha}{(1-\sin^2\alpha)(1+\sin^2\alpha)}\right] \times \cos^2\alpha \sin^2\alpha$$

$$= \left[\frac{\cos^2\alpha}{\sin^2\alpha(1+\cos^2\alpha)} + \frac{\sin^2\alpha}{\cos^2\alpha(1+\sin^2\alpha)}\right]\cos^2\alpha \sin^2\alpha$$

$$= \left[\frac{\cos^4\alpha(1+\sin^2\alpha) + \sin^4\alpha(1+\cos^2\alpha)}{\sin^2\alpha\cos^2\alpha(1+\cos^2\alpha)(1+\sin^2\alpha)}\right]\cos^2\alpha \sin^2\alpha$$

$$= \left[\frac{\cos^4\alpha + \cos^4\alpha \sin^2\alpha + \sin^4\alpha + \sin^4\alpha.\cos^2\alpha}{\left(1+\cos^2\alpha\right)\left(1+\sin^2\alpha\right)} \right]$$

$$= \frac{\left(\cos^2\alpha + \sin^2\alpha\right)^2 + \cos^2\alpha \sin^2\alpha\left(\sin^2\alpha + \cos^2\alpha\right) - 2\cos^2\alpha \sin^2\alpha}{1+\sin^2\alpha + \cos^2\alpha + \cos^2\alpha \sin^2\alpha}$$

$$= \frac{1+\cos^2\alpha \sin^2\alpha - 2\cos^2\alpha \sin^2\alpha}{2+\cos^2\alpha \sin^2\alpha}$$

$$= \frac{1-\cos^2\alpha \sin^2\alpha}{2+\cos^2\alpha \sin^2\alpha}$$

19. $\sin^8 A - \cos^8 A = (\sin^2 A - \cos^2 A)\,(1 - 2\sin^2 A \cos^2 A)$

Ans.

Sol. L.H.S. $= \left(\sin^8 A - \cos^8 A\right)$

$$= \left(\sin^4 A + \cos^4 A\right)\left(\sin^4 A - \cos^4 A\right)$$

$$= \left(\sin^4 A + \cos^4 A\right)\left(\sin^2 A - \cos^2 A\right)\left(\sin^2 A + \cos^2 A\right)$$

$$= \left[\left(\sin^2 A + \cos^2 A\right)^2 - 2\sin^2 A \cos^2 A\right]\left(\sin^2 A - \cos^2 A\right)$$

$$= \left(1 - 2\sin^2 A \cos^2 \mathbf{A}\right)\left(\sin^2 A - \cos^2 \mathbf{A}\right)$$

20. $\dfrac{\cos A \cos ecA - \sin A \sec A}{\cos A + \sin A} = \cos ecA - \sec A$

Ans.

Sol. L.H.S. $= \dfrac{\cos A \,\mathrm{cosec}\, A - \sin A \sec A}{\cos A + \sin A}$

$$= \frac{\cos A \times \dfrac{1}{\sin A} - \sin A \times \dfrac{1}{\cos A}}{\cos A + \sin A}$$

$$= \frac{\dfrac{\cos A}{\sin A} - \dfrac{\sin A}{\cos A}}{\cos A + \sin A} = \frac{\cos^2 A - \sin^2 A}{\sin A \cos A(\cos A + \sin A)}$$

$$= \frac{(\sin A + \cos A)(\cos A - \sin A)}{\sin A \cos A(\cos A + \sin A)} = \frac{\cos A - \sin A}{\sin A \cos A}$$

$$= \frac{1}{\sin A} - \frac{1}{\cos A}$$

$$= \operatorname{cosec} A - \sec A$$

21. $\dfrac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \dfrac{1 + \sin A}{\cos A}$

Ans.

Sol. L.H.S. $= \dfrac{\tan A + \sec A - 1}{\tan A - \sec A + 1}$

$$= \frac{(\tan A + \sec A) - \left(\sec^2 A - \tan^2 A\right)}{\tan A - \sec A + 1}$$

$$= \frac{(\tan A + \sec A)[1 - \sec A + \tan A]}{\tan A - \sec A + 1}$$

$$= \tan A + \sec A = \frac{\sin A}{\cos A} + \frac{1}{\cos A}$$

$$= \frac{\sin A + 1}{\cos A}$$

22. $\left(\tan \alpha + \cos ec \beta\right)^2 - (\cot \beta - \sec \alpha)^2 = 2 \tan \alpha \cot \beta (\operatorname{cosec} \alpha + \sec \beta)$

Ans.

Sol. L.H.S. $= (\tan \alpha + \operatorname{cosec} \beta)^2 - (\cot \beta - \sec \alpha)^2$

L.H.S. $= \left(\tan^2 \alpha + \operatorname{cosec}^2 \beta + 2 \tan \alpha \operatorname{cosec} \beta\right) - \left(\cot^2 \beta + \sec^2 \alpha - 2 \cot \beta \sec \alpha\right)$

$$= \tan^2 \alpha - \sec^2 \alpha + \operatorname{cosec}^2 \beta - \cot^2 \beta + 2(\tan \alpha \operatorname{cosec} \beta + \cot \beta \sec \alpha)$$

$$= \tan^2 \alpha - 1 - \tan^2 \alpha + 1 + \cot^2 \beta - \cot^2 \beta + 2 \tan \alpha \cot \beta \left[\frac{\operatorname{cosec} \beta}{\cot \beta} + \frac{\sec \alpha}{\tan \alpha}\right]$$

$$= 2 \tan \alpha \cot \beta \left[\frac{1}{\cos \beta} + \frac{1}{\sin \alpha}\right]$$

$$= 2 \tan \alpha \cot \beta (\sec \beta + \operatorname{cosec} \alpha)$$

23. $2\sec^2 \alpha - \sec^4 \alpha - 2\operatorname{cosec}^2 \alpha + \operatorname{cosec}^4 \alpha = \cot^4 \alpha - \tan^4 \alpha$

Ans.

Sol. L.H.S. $= 2\sec^2\alpha - \sec^4\alpha - 2\,\text{cosec}^2\alpha + \text{cosec}^4\alpha$

$= 2\sec^2\alpha - \sec^4\alpha - 1 + 1 - 2\,\text{cosec}^2\alpha + \text{cosec}^4\alpha$

$= \left(\text{cosec}^4\alpha - 2\,\text{cosec}^2\alpha + 1\right) - \left(\sec^4\alpha - 2\sec^2\alpha + 1\right)$

$= \left(\text{cosec}^2\alpha - 1\right)^2 - \left(\sec^2\alpha - 1\right)^2$

$= \left(1 + \cot^2\alpha - 1\right)^2 - \left(1 + \tan^2\alpha - 1\right)^2$

$= \left(\cot^2\alpha\right)^2 - \left(\tan^2\alpha\right)^2$

$= \cot^4\alpha - \tan^4\alpha$

24. $(\sin\alpha + \text{cosec}\,\alpha)^2 + (\cos\alpha + \sec\alpha)^2 = \tan^2\alpha + \cot^2\alpha + 7$

Ans.

Sol. L.H.S. $= (\sin\alpha + \text{cosec}\,\alpha)^2 + (\cos\alpha + \sec\alpha)^2$

$= \sin^2\alpha + \text{cosec}^2\alpha + 2\sin\alpha\,\text{cosec}\,\alpha + \cos^2\alpha + \sec^2\alpha + 2\cos\alpha\sec\alpha$

$= \left(\sin^2\alpha + \cos^2\alpha\right) + 1 + \cot^2\alpha + 2 + 1 + \tan^2\alpha + 2$

$= 1 + 6 + \tan^2\alpha + \cot^2\alpha$

$= \tan^2\alpha + \cot^2\alpha + 7$

25. $(1 + \cot A + \tan A)(\sin A - \cos A) = \dfrac{\sec A}{\text{cosec}^2 A} - \dfrac{\text{cosec}\,A}{\sec^2 A}$

Ans.

Sol. R.H.S $= \dfrac{\sec A}{\text{cosec}^2 A} - \dfrac{\text{cosec}\,A}{\sec^2 A} = \dfrac{\sec^3 A - \text{cosec}^3 A}{\text{cosec}^2 A \sec^2 A}$

$= \dfrac{\left(\sec A - \text{cosec}\,A\right)\left(\sec^2 A + \text{cosec}^2 A + \sec A\,\text{cosec}\,A\right)}{\text{cosec}^2 A \sec^2 A}$

$= \sin A - \cos A\left[\dfrac{\sec A}{\text{cosec}\,A} + \dfrac{\text{cosec}\ A}{\sec A} + 1\right]$

$= (\sin A - \cos A)(1 + \tan A + \cot A)$

Prove that

सिद्ध कीजिये

1. $\sin 420^\circ \cos 390^\circ + \cos(-300^\circ)\sin(-330^\circ) = 1$

Ans.

Sol. L.H.S. $= \sin(90° + 330°)\cos(90° + 300°) - \sin(330°)\cos 300°$

$= -[\cos 330° \sin(300°) + \sin 330° \cos 300°]$

$= -[\sin(300° + 330°] = 1$

2. $\cos 570^\circ \sin 510^\circ - \sin 330^\circ \cos 390^\circ = 0$

Ans.

Sol. L.H.S. $= \cos(180° + 390°)\sin(180° + 330°) - \sin 330° \cos 390°$

$= \cos(390°)\sin(330°) - \sin 330° \cos 390°$

$= 0$

3. $\tan 225^\circ \cot 405^\circ + \tan 765^\circ \cot 675^\circ = 0$

Ans.

Sol. L.H.S. $= \tan(180° + 45°)\cot(360° + 45°) + \tan(720° + 45°)\cot(720° - 45°)$

$= \tan 45° \cot 45° - \cot 45° \tan 45° = 0$

What are the values of $\cos A - \sin A$ and $\tan A + \cot A$ when A has the value of

$\cos A - \sin A$ तथा $\tan A + \cot A$ का मान क्या होगा जबकि A के मान नीचे दिये गये है।

4. $\frac{\pi}{3}$

Ans. –0.366....; 2.3094...

Sol. $\cos A - \sin A = \cos\frac{\pi}{3} - \sin\frac{\pi}{3} = \frac{1-\sqrt{3}}{2}$

$$\tan A + \cot A = \tan\frac{\pi}{3} + \cot\frac{\pi}{3} = \frac{4}{\sqrt{3}}$$

5. $\frac{2\pi}{3}$

Ans. –1.366....; –2.3094

Sol. $\cos\left(\frac{2\pi}{3}\right) - \sin\left(\frac{2\pi}{3}\right) = \frac{-\sqrt{3}-1}{2}$

$$\tan\left(\frac{2\pi}{3}\right) + \cot\left(\frac{2\pi}{3}\right) = -\frac{4}{\sqrt{3}}$$

6. $\frac{5\pi}{4}$

Ans. 0; 2

Sol. $\cos\frac{5\pi}{4} - \sin\frac{5\pi}{4} = -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = 0$

$$\tan\left(\frac{5\pi}{4}\right) + \cot\left(\frac{5\pi}{4}\right) = 1 + 1 = 2$$

7. $\frac{7\pi}{4}$

Ans. 1.4142.... –2

Sol. $\cos\left(\frac{7\pi}{4}\right) - \sin\left(\frac{7\pi}{4}\right) = \cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right) = \sqrt{2}$

$$\tan\left(\frac{7\pi}{4}\right) + \cot\left(\frac{7\pi}{4}\right) = -\tan\left(\frac{7\pi}{4}\right) - \cot\left(\frac{7\pi}{4}\right) = -2$$

8. $\frac{11\pi}{3}$

Ans. 1.366.....; –2.3094...

Sol. $\cos\left(\frac{11\pi}{3}\right)-\sin\left(\frac{11\pi}{3}\right)=\frac{1}{2}+\frac{\sqrt{3}}{2}=\frac{1+\sqrt{3}}{2}$

$\tan\left(\frac{11\pi}{3}\right)+\cot\left(\frac{11\pi}{3}\right)=\tan\left(4\pi-\frac{\pi}{3}\right)+\cot\left(4\pi-\frac{\pi}{3}\right)$

$=-\sqrt{3}-\frac{1}{\sqrt{3}}=-\frac{4}{\sqrt{3}}$

What values between 0º and 360º may A have when

0º तथा 360º के मध्य A के मान क्या होंगे जबकि

9. $\sin A=\frac{1}{\sqrt{2}}$

Ans. 45º and 135º

Sol. $\sin A=\frac{1}{\sqrt{2}}\Rightarrow A=\frac{\pi}{4}\equiv 45^\circ$ or $\pi-\frac{\pi}{4}=135^\circ$

10. $\cos A=-\frac{1}{2}$

Ans. 120º and 240º

Sol. $\cos A=-\frac{1}{2}\Rightarrow A=\pi-\frac{\pi}{3},\pi+\frac{\pi}{3}$

11. $\tan A=-1$

Ans. 135º and 315º

Sol. $\tan A=-1\Rightarrow A=\pi-\frac{\pi}{4},2\pi-\frac{\pi}{4}$

12. $\cot A = -\sqrt{3}$

Ans. 150º and 330º

Sol. $\cot A = -\sqrt{3} \Rightarrow A = \pi - \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$

13. $\sec A = -\frac{2}{\sqrt{3}}$

Ans. 150º and 210º

Sol. $\sec A = -\frac{2}{\sqrt{3}} \Rightarrow A = \pi - \frac{\pi}{6}, \pi + \frac{\pi}{6}$

14. $\text{cosec } A = -2$

Ans. 210º and 330º

Sol. $\text{cosec } A = -2 \Rightarrow A = \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$

Express in terms of the ratios of a positive angle, which is less than 45º, the quantities

नीचे दिये गये अनुपातों को धनात्मक तथा 45º से छोटे कोणों में व्यक्त करो।

15. $\sin(-65^\circ)$

Ans. $-\cos 25^\circ$

Sol. $\sin(-65^\circ) = -\sin(90^\circ - 25^\circ) = -\cos 25^\circ$

16. $\cos(-84^\circ)$

Ans. $\sin 6^\circ$

Sol. $\cos(-84^\circ) = \cos(90^\circ - 6^\circ) = \sin 6^\circ$

17. $\tan 137^\circ$

Ans. $-\tan 43^\circ$

Sol. $\tan(137^\circ) = \tan(180^\circ - 43^\circ) = -\tan 43^\circ$

18. $\sin 168^\circ$

Ans. $\sin 12^\circ$

Sol. $\sin(168^\circ) = \sin(180^\circ - 12^\circ) = \sin(12^\circ)$

19. $\cos 287^\circ$

Ans. $\sin 17^\circ$

Sol. $\cos(287^\circ) = \cos(270^\circ + 17^\circ) = \sin(17^\circ)$

20. $\tan(-246^\circ)$

Ans. $-\cot 24^\circ$

Sol. $\tan(-246^\circ) = -\tan(246^\circ) = -\tan(270^\circ - 24^\circ) = -\cot(24^\circ)$

21. $\sin 843^\circ$

Ans. $\cos 33^\circ$

Sol. $\sin(843^\circ) = \sin(810^\circ + 33^\circ) = \cos 33^\circ$

22. $\cos(-928^\circ)$

Ans. $-\cos 28^\circ$

Sol. $\cos(-928^\circ) = \cos(928^\circ) = \cos(900^\circ + 28^\circ) = -\cos 28^\circ$

23. $\tan 1145^\circ$

Ans. $\cot 25^\circ$

Sol. $\tan(1145^\circ) = \tan(1170^\circ - 25^\circ) = \cot(25^\circ)$

24. $\cos 1410^\circ$

Ans. $\cos 30^\circ$

Sol. $\cos(1410^\circ) = \cos(1440^\circ - 30^\circ) = \cos 30^\circ$

25. cot (–1054º)

Ans. cot 26º

Sol. cot(–1054°) = –cot (1054°) = –cot (1080° – 26°) = cot (26°)

26. sec 1327º

Ans. –cosec 23º

Sol. sec(1327°) = sec (1350° – 23°) = –cosec 23°

27. cosec (–756º)

Ans. –cosec 36º

Sol. cosec (–756°) = –cosec (720° + 36°) = –cosec 36°

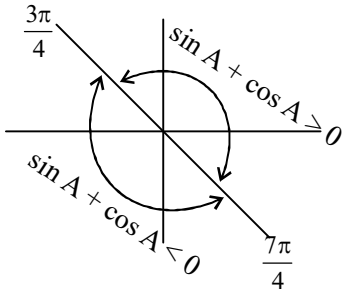
What sign has sinA + cos A for the following values of A ?

निम्न A के मानों के लिये sinA + cos A का चिन्ह कया होगा ?

28. 140º

Ans. Negative

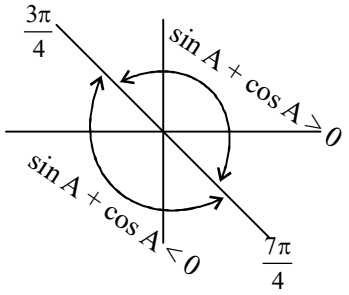
Sol.



29. 278º

Ans. Negative

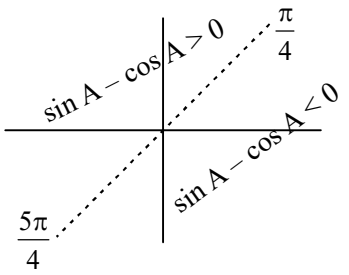
Sol.



30. –356º

Ans. Positive

Sol. cos (356°) – sin (356°) is positive in IVth Quadrant.



31. –1125º

Ans. Zero

Sol. cos (–1125°) + sin (–1125°) = cos 45° – sin 45° = 0

What sign has sin A – cos A for the following values of A ?

निम्न A के मानों के लिये sin A – cos A का चिन्ह कया होगा ?

32. 215º

Ans. Positive

Sol.

33. 825º

Ans. Positive

Sol.

34. –634º

1. If $\sin\alpha = \frac{3}{5}$ and $\cos\beta = \frac{9}{41}$, find the value of $\sin(\alpha - \beta)$ and $\cos(\alpha + \beta)$. (α, β are acute angle)

यदि $\sin\alpha = \frac{3}{5}$ तथा $\cos\beta = \frac{9}{41}$ तो $\sin(\alpha - \beta)$ तथा $\cos(\alpha + \beta)$ का मान ज्ञात करो

(α, β न्यून कोण है)

Ans. $-\frac{133}{205}; -\frac{84}{205}$

Sol. $\sin\alpha = \frac{3}{5}$

We know, $\cos\alpha = \sqrt{(1-\sin^2\alpha)} = \sqrt{\left(1-\frac{9}{25}\right)} = \frac{4}{5}$

and $\cos\beta = \frac{9}{41}$

$\therefore$ $\sin\beta = \sqrt{(1-\cos^2\beta)} = \sqrt{\left(1-\frac{81}{1681}\right)} = \frac{40}{41}$

Now $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \sin\beta\cos\alpha.$

Putting the values, we have

$$\sin(\alpha-\beta) = \frac{3}{5}\times\frac{9}{41} - \frac{40}{41}\times\frac{4}{5} = \frac{27-160}{205} = -\frac{133}{205}$$

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

$$= \frac{4}{5}\times\frac{9}{41} - \frac{3}{5}\times\frac{40}{41}$$

$$= \frac{36-120}{205} = -\frac{84}{205}$$

2. If $\sin\alpha = \frac{15}{17}$ and $\cos\beta = \frac{12}{13}$, find the values of $\sin(\alpha + \beta)$, $\cos(\alpha - \beta)$ and $\tan(\alpha + \beta)$. (α, β are acute angle)

यदि $\sin\alpha = \frac{15}{17}$ तथा $\cos\beta = \frac{12}{13}$, तो $\sin(\alpha + \beta)$, $\cos(\alpha - \beta)$ तथा $\tan(\alpha + \beta)$ का मान ज्ञात करो

(α, β एक न्यून कोण हैं)

Ans. $\frac{220}{221};\frac{171}{221};\frac{220}{21}$

Sol. Given, $\sin\alpha=\frac{15}{17}$

$\therefore$ $\cos\alpha=\sqrt{\left\{1-\left(\frac{15}{17}\right)^2\right\}}=\frac{8}{17}$

and $\cos\beta=\frac{12}{13}$ (given)

$\therefore$ $\sin\beta=\sqrt{\left\{1-\left(\frac{12}{13}\right)^2\right\}}=\frac{5}{13}$

Now, $\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$

$$=\frac{15}{17}\cdot\frac{12}{13}+\frac{5}{13}\times\frac{8}{17}=\frac{220}{221}$$

and $\cos(\alpha-\beta)=\cos\alpha\cos\beta+\sin\alpha\sin\beta$

$$=\frac{8}{17}\times\frac{12}{13}+\frac{15}{17}\times\frac{5}{13}=\frac{171}{221}$$

$\therefore$ $\tan(\alpha+\beta)=\frac{\sin(\alpha+\beta)}{\cos(\alpha+\beta)}=\frac{\sin(\alpha+\beta)}{\cos\alpha\cos\beta-\sin\alpha\sin\beta}$

$$=\frac{\sin(\alpha+\beta)}{\frac{8}{17}\times\frac{12}{13}-\frac{15}{17}\times\frac{5}{13}}=\frac{\frac{220}{221}}{\frac{21}{221}}=\frac{220}{21}$$

3. Prove that $\cos(45^\circ-A)\cos(45^\circ-B)-\sin(45^\circ-A)\sin(45^\circ-B)=\sin(A+B)$

सिद्ध कीजिये $\cos(45^\circ-A)\cos(45^\circ-B)-\sin(45^\circ-A)\sin(45^\circ-B)=\sin(A+B)$

Sol. L.H.S. $=\cos\left(45^\circ-A+45^\circ-B\right)$

$=\cos\left(90^\circ-A-B\right)=\cos\left[90^\circ-(A+B)\right.$

$=\sin(A+B)$

Prove that

सिद्ध कीजिये

4. $\sin(45^\circ + A)\cos(45^\circ - B) + \cos(45^\circ + A)\sin(45^\circ - B) = \cos(A - B)$

Ans.

Sol. L.H.S. $= \sin\left(45^\circ + A + 45^\circ - B\right) = \sin\left[90^\circ + (A - B)\right] = \cos(A - B)$

$= \cos(A - B)$

5. $\dfrac{\sin(A-B)}{\cos A\cos B} + \dfrac{\sin(B-C)}{\cos B\cos C} + \dfrac{\sin(C-A)}{\cos C\cos A} = 0$

Ans.

Sol. L.H.S. $= \dfrac{\sin(A-B)}{\cos A\cos B} + \dfrac{\sin(B-C)}{\cos B\cos C} + \dfrac{\sin(C-A)}{\cos C\cos A}$

$= \dfrac{\sin A\cos B - \cos A\sin B}{\cos A\cos B} + \dfrac{\sin B\cos C - \sin C\cos B}{\cos B\cos C}$

$+ \dfrac{\sin C\cos A - \cos C\sin A}{\cos A\cos C}$

$= \dfrac{\sin A\cos B}{\cos A\cos B} - \dfrac{\cos A\sin B}{\cos A\cos B} + \dfrac{\sin B\cos C}{\cos B\cos C} - \dfrac{\sin C\cos B}{\cos B\cos C} + \dfrac{\sin C\cos A}{\cos A\cos C} - \dfrac{\cos C\sin A}{\cos A\cos C}$

$= \tan A - \tan B + \tan B - \tan C + \tan C - \tan A$

$= 0$

6. $\sin 105^\circ + \cos 105^\circ = \cos 45^\circ$

Ans.

Sol. L.H.S. $= \sin 105^\circ + \cos 105^\circ$

$= \sin\left(60^\circ + 45^\circ\right) + \cos\left(60^\circ + 45^\circ\right)$

$= \sin 60^\circ\cos 45^\circ + ^\circ\cos 60^\circ\sin 45^\circ + \cos 60^\circ\cos 45^\circ - \sin 60^\circ\sin 45^\circ$

$= \dfrac{\sqrt{3}}{2}\cdot\dfrac{1}{\sqrt{2}} + \dfrac{1}{2}\cdot\dfrac{1}{\sqrt{2}} + \dfrac{1}{2}\cdot\dfrac{1}{\sqrt{2}} - \dfrac{\sqrt{3}}{2}\cdot\dfrac{1}{\sqrt{2}}$

$= 2\cdot\dfrac{1}{2}\cdot\dfrac{1}{\sqrt{2}} = \dfrac{1}{\sqrt{2}} = \cos 45^\circ$

7. $\sin 75^\circ - \sin 15^\circ = \cos 105^\circ + \cos 15^\circ$

Ans.

Sol. R.H.S. $= \cos\left(90^\circ + 15^\circ\right) + \cos\left(90^\circ - 75^\circ\right)$

$= -\sin 15^\circ + \sin 75^\circ$

$= \sin 75^\circ - \sin 15^\circ$

8. $\cos\alpha \cos(\gamma - \alpha) - \sin\alpha \sin(\gamma - \alpha) = \cos\gamma$

Ans.

Sol. L.H.S. $= \cos\alpha\cos(\gamma - \alpha) - \sin\alpha\sin(\gamma - \alpha)$

$= \cos(\alpha + \gamma - \alpha) = \cos\gamma$

$=$ R.H.S.

9. $\cos(\alpha + \beta)\cos\gamma - \cos(\beta + \gamma)\cos\alpha = \sin\beta \sin(\gamma - \alpha)$

Ans.

Sol. L.H.S. $= \cos(\alpha + \beta)\cos\gamma - \cos(\beta + \gamma)\cos\alpha$

$= \cos\gamma(\cos\alpha\cos\beta - \sin\alpha\sin\beta) - (\cos\beta\cos\gamma - \sin\beta\sin\gamma)\cos\alpha$

$= \cos\gamma\cos\alpha\cos\beta - \sin\alpha\sin\beta\cos\gamma - \cos\alpha\cos\beta\cos\gamma + \cos\alpha\sin\beta\sin\gamma$

$= \cos\alpha\sin\beta\sin\gamma - \sin\alpha\sin\beta\cos\gamma$

$= \sin\beta(\cos\alpha\sin\gamma - \sin\alpha\cos\gamma)$

$= \sin\beta\sin(\gamma - \alpha)$

10. $\sin(n + 1)A \sin(n - 1)A + \cos(n + 1)A \cos(n - 1)A = \cos 2A$

Ans.

Sol. L.H.S. $= \cos(n+1)A\cos(n-1)A + \sin(n+1)A\sin(n-1)A$

$= \cos\{(n+1)A - (n-1)A\}$

$= \cos\{nA + A - nA + A\}$

$= \cos 2A$

11. $\sin(n + 1)A \sin(n + 2)A + \cos(n + 1)A \cos(n + 2)A = \cos A$

Ans.

Sol. L.H.S. $= \cos(n+1)A\cos(n+2)A + \sin(n+1)A\sin(n+2)A$

$= \cos\{(n+2)A - (n+1)A\}$

$$= \cos\{An + 2A - An - A\}$$
$$= \cos A$$

12. $\dfrac{\sin 7\theta - \sin 5\theta}{\cos 7\theta + \cos 5\theta} = \tan\theta$

Ans.

Sol. $\dfrac{\sin 7\theta - \sin 5\theta}{\cos 7\theta + \cos 5\theta} = \dfrac{2\cos 6\theta \sin\theta}{2\cos 6\theta \cos\theta}$

$$= \frac{\sin\theta}{\cos\theta} = \tan\theta$$

13. $\dfrac{\cos 6\theta - \cos 4\theta}{\sin 6\theta + \sin 4\theta} = -\tan\theta$

Ans.

Sol. $\dfrac{\cos 6\theta - \cos 4\theta}{\sin 6\theta + \sin\theta} = \dfrac{\cos 6\theta - \cos 4\theta}{\sin 6\theta + \sin 4\theta} = \dfrac{-2\sin 5\theta \sin\theta}{2\sin 5\theta \cos\theta}$

$$= -\frac{\sin\theta}{\cos\theta} = -\tan\theta$$

14. $\dfrac{\sin A + \sin 3A}{\cos A + \cos 3A} = \tan 2A$

Ans.

Sol. $\dfrac{\sin A + \sin 3A}{\cos A + \cos 3A} = \dfrac{2\sin 2A \cos A}{2\cos 2A \cos A}$

$$= \frac{\sin 2A}{\cos 2A} = \tan 2A$$

15. $\dfrac{\cos 2B + \cos 2A}{\cos 2B - \cos 2A} = \cot(A+B)\cot(A-B)$

Ans.

Sol. $\dfrac{\cos 2B+\cos 2A}{\cos 2B-\cos 2A}=\dfrac{2\cos(A+B)\cos(A-B)}{2\sin(A+B)\sin(A-B)}$

$$=\cot(A+B)\cot(A-B)$$

16. $\dfrac{\sin 2A+\sin 2B}{\sin 2A-\sin 2B}=\dfrac{\tan(A+B)}{\tan(A-B)}$

Ans.

Sol. $\dfrac{\sin 2A+\sin 2B}{\sin 2A-\sin 2B}=\dfrac{2\sin(A+B)\cos(A-B)}{2\cos(A+B)\sin(A-B)}$

$$=\tan(A+B)\cot(A-B)=\frac{\tan(A+B)}{\tan(\mathbf{A-B})}$$

17. $\dfrac{\sin A+\sin 2A}{\cos A-\cos 2A}=\cot\dfrac{A}{2}$

Ans.

Sol. $\dfrac{\sin A+\sin 2A}{\cos A-\cos 2A}=\dfrac{2\sin\dfrac{3A}{2}\cos\dfrac{A}{2}}{2\sin\dfrac{3A}{2}\sin\dfrac{A}{2}}$

$=\cot\dfrac{A}{2}$

18. $\cos(A+B)+\sin(A-B)=2\sin(45^o+A)\cos(45^o+B)$

Ans.

Sol. $\cos(A+B)+\sin(A-B)=\cos(A+B)+\sin(A-B)$

$$=\sin(90^\circ+(A+B)\}+\sin(A-B)\left[\because \sin\left(90^\circ+\theta\right)=\cos\theta\right]$$

$$=2\sin\left(\frac{90^\circ+2A}{2}\right)\cos\left(\frac{90^\circ+2B}{2}\right)$$

$$=2\sin\left(45^\circ+\mathbf{A}\right)\cos\left(45^\circ+B\right)$$

19. $\dfrac{\cos 3A-\cos A}{\sin 3A-\sin A}+\dfrac{\cos 2A-\cos 4A}{\sin 4A-\sin 2A}=\dfrac{\sin A}{\cos 2A\cos 3A}$

Ans.

Sol. $\dfrac{\cos 3A-\cos A}{\sin 3A-\sin A}+\dfrac{\cos 2A-\cos 4A}{\sin 4A-\sin 2A}$

$$=\frac{2\sin 2A\sin(-A)}{2\cos 2A\sin A}+\frac{2\sin 3A\sin A}{2\cos 3A\sin A}$$

$$=-\frac{\sin 2A}{\cos 2A}+\frac{\sin 3A}{\cos 3A}=\frac{\sin 3A}{\cos 3A}-\frac{\sin 2A}{\cos 2A}$$

$$=\frac{\sin 3A\cos 2A-\sin 2A\cos 3A}{\cos 2A\cos 3A}$$

$$=\frac{\sin(3A-2A)}{\cos 2A\cos 3A}$$

$$=\frac{\sin A}{\cos 2A\cos 3A}$$

20. $\dfrac{\tan 5\theta+\tan 3\theta}{\tan 5\theta-\tan 3\theta}=4\cos 2\theta\cos 4\theta$

Ans.

Sol. L.H.S. $=\dfrac{\dfrac{\sin 5\theta}{\cos 5\theta}+\dfrac{\sin 3\theta}{\cos 3\theta}}{\dfrac{\sin 5\theta}{\cos 5\theta}-\dfrac{\sin 3\theta}{\cos 3\theta}}=\dfrac{\dfrac{\sin 5\theta\cos 3\theta+\cos 5\theta\sin 3\theta}{\cos 5\theta\cos 3\theta}}{\dfrac{\sin 5\theta\cos 3\theta-\cos 5\theta\sin 3\theta}{\cos 5\theta\cos 3\theta}}$

$$=\frac{\sin(5\theta+3\theta)}{\sin(5\theta-3\theta)}=\frac{\sin 8\theta}{\sin 2\theta}=\frac{2\sin 4\theta\cos 4\theta}{\sin 2\theta}$$

$$=\frac{2\times 2\sin 2\theta\cos 2\theta\cos 4\theta}{\sin 2\theta}$$

$$=4\cos 2\theta\cos 4\theta.$$

21. $\dfrac{\cos 3\theta+2\cos 5\theta+\cos 7\theta}{\cos\theta+2\cos 3\theta+\cos 5\theta}=\cos 2\theta-\sin 2\theta\tan 3\theta$

Ans.

Sol. $\dfrac{(\cos 3\theta + \cos 7\theta) + 2\cos 5\theta}{(\cos\theta + \cos 5\theta) + 2\cos 3\theta}$

$$= \frac{2\cos 5\theta\cos 2\theta + 2\cos 5\theta}{2\cos 3\theta\cos 2\theta + 2\cos 3\theta}$$

$$= \frac{\cos(3\theta + 2\theta)}{\cos 3\theta} = \frac{\cos 3\theta\cos 2\theta - \sin 2\theta\sin 3\theta}{\cos 3\theta}$$

$$= \frac{\cos 3\theta\cos 2\theta}{\cos 3\theta} - \frac{\sin 2\theta\sin 3\theta}{\cos 3\theta}$$

$$= \cos 2\theta - \sin 2\theta\tan 3\theta$$

22. $\dfrac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A} = \tan 4A$

Ans.

Sol. $\dfrac{\sin A + \sin 3A + \sin 5A + \sin 7A}{\cos A + \cos 3A + \cos 5A + \cos 7A}$

$$= \frac{2\sin 2A\cos A + 2\sin 6A\cos A}{2\cos 2A\cos A + 2\cos 6A\cos A}$$

$$= \frac{2\cos A(\sin 2A + \sin 6A)}{2\cos A(\cos 2A + \cos 6A)}$$

$$= \frac{\sin 2A + \sin 6A}{\cos 2A + \cos 6A}$$

$$= \frac{2\sin 4A\cos 2A}{2\cos 4A\cos 2A}$$

$$= \frac{\sin 4A}{\cos 4A}$$

$$= \tan 4A$$

23. $\dfrac{\sin(\theta + \phi) - 2\sin\theta + \sin(\theta - \phi)}{\cos(\theta + \phi) - 2\cos\theta + \cos(\theta - \phi)} = \tan\theta$

Ans.

Sol. $\dfrac{\sin(\theta + \phi) + \sin(\theta - \phi) - 2\sin\theta}{\cos(\theta + \phi) + \cos(\theta - \phi) - 2\cos\phi}$

$$= \frac{2\sin\theta\cos\phi - 2\sin\theta}{2\cos\theta\cos\phi - 2\cos\theta}$$

$$= \frac{2\sin\theta(\cos\phi - 1)}{2\cos\theta(\cos\phi - 1)}$$

$$= \frac{\sin\theta}{\cos\theta}$$

$$= \tan\theta$$

24. $\dfrac{\sin A + 2\sin 3A + \sin 5A}{\sin 3A + 2\sin 5A + \sin 7A} = \dfrac{\sin 3A}{\sin 5A}$

Ans.

Sol. $\dfrac{\sin A + \sin 5A + 2\sin 3A}{\sin 3A + \sin 7A + 2\sin 5A}$

$$= \frac{2\sin 3A\cos 2A + 2\sin 3A}{2\sin 5A\cos 2A + 2\sin 5A}$$

$$= \frac{2\sin 3A(\cos 2A + 1)}{2\sin 5A(\cos 2A + 1)}$$

$$= \frac{\sin 3A}{\sin 5A}$$

25. $\dfrac{\sin(A-C) + 2\sin A + \sin(A+C)}{\sin(B-C) + 2\sin B + \sin(B+C)} = \dfrac{\sin A}{\sin B}$

Ans.

Sol. $\dfrac{\sin(A-C) + \sin(A+C) + 2\sin A}{\sin(B-C) + \sin(B+C) + 2\sin B}$

$$= \frac{2\sin A\cos C + 2\sin A}{2\sin B\cos C + 2\sin B} = \frac{2\sin A(\cos C + 1)}{2\sin B(\cos C + 1)}$$

$$= \frac{\sin A}{\sin B}$$

26. $\dfrac{\sin A - \sin 5A + \sin 9A - \sin 13A}{\cos A - \cos 5A - \cos 9A + \cos 13A} = \cot 4A$

Ans.

Sol. $\dfrac{(\sin A-\sin 5A)+(\sin 9A-\sin 13A)}{(\cos A-\cos 5A)-(\cos 9A-\cos 13A)}$

$$=\frac{2\sin(-2A)\{\cos 3A+\cos 11A\}}{2\sin 2A\{\sin 3A-\sin 11A\}}$$

$$=-\left(\frac{\cos 3A+\cos 11A}{\sin 3A-\sin 11A}\right)$$

$$=\frac{\cos 3A+\cos 11A}{\sin 11A-\sin 3A}$$

$$=\frac{2\cos 7A\cos 4A}{2\cos 7A\sin 4A}$$

$$=\frac{\cos 4A}{\sin 4A}$$

$$=\cot 4A$$

27. $\dfrac{\cos A+\cos B}{\cos B-\cos A}=\cot\dfrac{A+B}{2}\cot\dfrac{A-B}{2}$

Ans.

Sol. $\dfrac{\cos A+\cos B}{\cos B-\cos A}$

$$=\frac{2\cos\dfrac{A+B}{2}\cos\dfrac{A-B}{2}}{2\sin\dfrac{A+B}{2}\sin\dfrac{A-B}{2}}$$

$$=\cot\frac{A+B}{2}\cot\frac{A-B}{2}$$

28. $\dfrac{\sin A-\sin B}{\cos B-\cos A}=\cot\dfrac{A+B}{2}$

Ans.

Sol. $\dfrac{\sin A-\sin B}{\cos B-\cos A}=\dfrac{2\cos\dfrac{A+B}{2}\sin\dfrac{A-B}{2}}{2\sin\dfrac{A+B}{2}\sin\dfrac{A-B}{2}}$

$$= \frac{\cos\dfrac{A+B}{2}}{\sin\dfrac{A+B}{2}} = \cot\frac{A+B}{2}$$

29. $$\frac{\cos(A+B+C)+\cos(-A+B+C)+\cos(A-B+C)+\cos(A+B-C)}{\sin(A+B+C)+\sin(-A+B+C)-\sin(A-B+C)+\sin(A+B-C)} = \cot B$$

Ans.

Sol. $$\frac{\cos(A+B+C)+\cos(-A+B+C)+\cos(A-B+C)+\cos(A+B-C)}{\sin(A+B+C)+\sin(-A+B+C)-\sin(A-B+C)+\sin(A+B-C)}$$

$$= \frac{2\cos(B+C)\cos A + 2\cos A\cos(C-B)}{2\sin(B+C)\cos A - 2\cos A\sin(C-B)}$$

$$= \frac{2\cos A\{\cos(B+C)+\cos(C-B)\}}{2\cos A\{\sin(B+C)-\sin(C-B)\}}$$

$$= \frac{2\cos B\cos C}{2\cos C\sin B}$$

$$= \frac{\cos B}{\sin B}$$

$$= \cot B$$

30. $\cos 3A + \cos 5A + \cos 7A + \cos 15A = 4\cos 4A\cos 5A\cos 6A$

Ans.

Sol. $(\cos 3A + \cos 5A) + (\cos 7A + \cos 15A)$

$$= 2\cos 4A\cos A + 2\cos 11A\cos 4A$$

$$= 2\cos 4A[\cos A + \cos 11A]$$

$$= 2\cos 4A\{2\cos 6A\cos 5A\}$$

$$= 4\cos 4A\cos 5A\cos 6A$$

31. $\cos(-A+B+C) + \cos(A-B+C) + \cos(A+B-C) + \cos(A+B+C) = 4\cos A\cos B\cos C$

Ans.

Sol. $2\cos\frac{(A-B+C-A+B+C)}{2}\cdot\cos\frac{(A-B+C+A-B-C)}{2}$

$+2\cos\frac{A+B-C+A+B+C}{2}\cos\frac{A+B-C-A-B-C}{2}$

$=2\cos C\cos(A-B)+2\cos(A+B)\cos(-C)$

$=2\cos C\{\cos(A-B)+\cos(A+B)\}$

$=2\cos C\{2\cos A\cos B\}$

$=4\cos A\cos B\cos C$

32. $\sin 50^o-\sin 70^o+\sin 10^o=0$

Ans.

Sol. $\sin 50^\circ-\sin 70^\circ+\sin 10^\circ$

$=2\cos 60^\circ\sin\left(-10^\circ\right)+\sin 10^\circ$

$=-2\times\frac{1}{2}\sin 10^\circ+\sin 10^\circ$

$=-\sin 10^\circ+\sin 10^\circ$

$=0$

33. $\sin 10^o+\sin 20^o+\sin 40^o+\sin 50^o=\sin 70^o+\sin 80^o$

Ans.

Sol. $\sin 50^\circ+\sin 10^\circ+\sin 20^\circ+\sin 40^\circ$

$=2\sin 30^\circ\cos 20^\circ+2\sin 30^\circ\cos 10^\circ$

$=2\sin^\circ 30^\circ\left\{\cos 20^\circ+\cos 10^\circ\right\}$

$=2\times\frac{1}{2}\left\{\cos\left(90^\circ-70^\circ\right)+\cos\left(90^\circ-80^\circ\right)\right\}$

$=\sin 70^\circ+\sin 80^\circ$

34. $\sin\alpha+\sin 2\alpha+\sin 4\alpha+\sin 5\alpha=4\cos\frac{\alpha}{2}\cos\frac{3\alpha}{2}\sin 3\alpha$

Ans.

Sol. $(\sin\alpha+\sin 5\alpha)+(\sin 2\alpha+\sin 4\alpha)$

$$=2\sin 3\alpha\cos 2\alpha+2\sin 3\alpha\cos\alpha$$

$$=2\sin 3\alpha(\cos 2\alpha+\cos\alpha)$$

$$=2\sin 3\alpha\left\{2\cos\frac{3\alpha}{2}\cos\frac{\alpha}{2}\right\}$$

$$=4\sin 3\alpha\cos\frac{3\alpha}{2}\cos\frac{\alpha}{2}$$

35. Simplify: $\cos\left\{\theta+\left(n-\frac{3}{2}\right)\phi\right\}-\cos\left\{\theta+\left(n+\frac{3}{2}\right)\phi\right\}$

सरल करो : $\cos\left\{\theta+\left(n-\frac{3}{2}\right)\phi\right\}-\cos\left\{\theta+\left(n+\frac{3}{2}\right)\phi\right\}$

Ans. $2\sin(\theta+n\phi)\sin\frac{3\phi}{2}$

Sol. $=2\sin\left\{\frac{\theta+(n-3/2)\phi\}+\theta+\{n+3/2)\phi}{2}\right\}\sin\left\{\frac{\theta+(n+3/2)\phi-\theta-(n-3/2)\phi}{2}\right\}$

$$=2\sin(\theta+n\phi)\sin\frac{3}{2}\phi$$

1. Find the value of $\sin 2\alpha$ when :

   $\sin 2\alpha$ का मान ज्ञात करो जबकि

   (i) $\cos\alpha = \frac{3}{5}$, (ii) $\sin\alpha = \frac{12}{13}$ (iii) $\tan\alpha = \frac{16}{63}$

Ans. (i) $\pm\frac{24}{25}$ (ii) $\pm\frac{120}{169}$ (iii) $\frac{2016}{4225}$

Sol. (i) $\cos\alpha = \frac{3}{5}$

$$\sin\alpha = \sqrt{\left(1-\cos^2\alpha\right)}$$

$$= \sqrt{\left\{1-\left(\frac{9}{25}\right)\right\}} = \pm\frac{4}{5}$$

$$\sin 2\alpha = 2\sin\alpha\cos\alpha$$

$$= \pm 2\times\frac{3}{5}\times\frac{4}{5} = \pm\frac{24}{25}$$

(ii) $\cos\alpha = \sqrt{\left(1-\sin^2\alpha\right)} = \sqrt{\left(1-\frac{144}{169}\right)} = \pm\frac{5}{13}$

$$\sin 2\alpha = 2\sin\alpha\cos\alpha = \pm 2\cdot\frac{12}{13}\cdot\frac{5}{12} = \pm\frac{120}{169}$$

(iii) $\sin 2\alpha = \frac{2\tan\alpha}{1+\tan^2\alpha} = \frac{2\times\frac{16}{63}}{1+\left(\frac{16}{63}\right)^2} = \frac{2016}{4225}$

2. Find the value of $\cos 2\alpha$ when :

   $\cos 2\alpha$ का मान ज्ञात करो जबकि

   (1) $\cos\alpha = \frac{15}{17}$, (2) $\sin\alpha = \frac{4}{5}$ (3) $\tan\alpha = \frac{5}{12}$

Ans. (1) $\frac{161}{289}$, (2) $\frac{-7}{25}$ (3) $\frac{119}{169}$

Sol. (i) $\cos 2\alpha = 2\cos^2\alpha - 1 = 2\left(\frac{15}{17}\right)^2 - 1 = \frac{161}{289}$

(ii) $\cos 2\alpha = 1 - 2\sin^2\alpha = 1 - 2\left(\frac{4}{5}\right)^2 = -\frac{7}{25}$

(iii) $\cos 2\alpha = \frac{1-\tan^2\alpha}{1+\tan^2\alpha} = \frac{1-\left(\frac{5}{12}\right)^2}{1+\left(\frac{5}{12}\right)^2} = \frac{119}{169}$

3. If $\tan\theta = \frac{b}{a}$, find the value of $a\cos 2\theta + b\sin 2\theta$

यदि $\tan\theta = \frac{b}{a}$, तो $a\cos 2\theta + b\sin 2\theta$ का मान ज्ञात करो।

Ans. a

Sol. $a\cos 2\theta + b\sin 2\theta = a\frac{1-\tan^2\theta}{1+\tan^2\theta} + b\frac{2\tan\theta}{1+\tan^2\theta}$

$= a\frac{1-b^2/a^2}{1+b^2/a^2} + b\frac{2\times b/a}{1+b^2/a^2}$

$= a\frac{a^2-b^2}{a^2+b^2} + b\frac{2ab}{a^2+b^2} = \frac{a^3-ab^2+2ab^2}{a^2+b^2}$

$= \frac{a^3+ab^2}{a^2+b^2} = \frac{a\left(a^2+b^2\right)}{a^2+b^2}$

$= a$

Prove that

सिद्ध कीजिये :

4. $\frac{\sin 2A}{1+\cos 2A} = \tan A$

Ans.

Sol. $\frac{\sin 2A}{1+\cos 2A} = \frac{2\sin A\cos A}{1+2\cos^2 A-1} = \frac{2\sin A\cos A}{2\cos^2 A}$

$= \tan A$

5. $\frac{\sin 2A}{1-\cos 2A} = \cot A$

Ans.

Sol. $LHS = \frac{\sin 2A}{1-\cos 2A} = \frac{2\sin A\cos A}{1-1+2\sin^2 A}$

$= \frac{2\sin A\cos A}{2\sin^2 A}$

$= \cot A$

6. $\frac{1-\cos 2A}{1+\cos 2A} = \tan^2 A$

Ans.

Sol. $\frac{1-\cos 2A}{1+\cos 2A} = \frac{1+2\sin^2 A-1}{1+2\cos^2 A-1} = \frac{2\sin^2 A}{2\cos^2 A}$

$= \tan^2 A$

7. $\tan A + \cot A = 2\ \text{cosec}\ 2A$

Ans.

Sol. $\tan A + \cot A = \frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} = \frac{\sin^2 A+\cos^2 A}{\sin A\cos A}$

$= \frac{2}{\sin 2A}$

$= 2\,\text{cosec}\,2A$

8. $\tan A - \cot A = -2\cot 2A$

Ans.

Sol. $\tan A - \cot A = \frac{\sin A}{\cos A} - \frac{\cos A}{\sin A} = \frac{\sin^2 A - \cos^2 A}{\sin A \cos A}$

$= \frac{-\left(\cos^2 A - \sin^2 A\right)}{\sin A \cos A} = -\frac{2\cos 2A}{2\sin A \cos A}$

$= -2\cot 2A$

9. $\text{cosec } 2A + \cot 2A = \cot A$

Ans.

Sol. $\text{cosec } 2A + \cot 2A = \frac{1}{\sin 2A} + \frac{\cos 2A}{\sin 2A} = \frac{1 + \cos 2A}{\sin 2A}$

$= \frac{1 + 2\cos^2 A - 1}{2\sin A \cos A} = \frac{\cos A}{\sin A}$

$= \cot A$

10. $\frac{1 - \cos A + \cos B - \cos(A+B)}{1 + \cos A - \cos B - \cos(A+B)} = \tan\frac{A}{2}\cot\frac{B}{2}$

Ans.

Sol. $\frac{1 - \cos A + \cos B - \cos(A+B)}{1 + \cos A - \cos B - \cos(A+B)}$

$= \frac{1 - 1 + 2\sin^2\frac{A}{2} + 2\sin\left(\frac{A}{2} + B\right)\sin\frac{A}{2}}{1 + 2\cos^2\frac{A}{2} - 1 - \left\{2\cos\left(\frac{A}{2} + B\right)\cos\frac{A}{2}\right\}}$

$= \frac{2\sin\frac{A}{2}\left\{\sin\frac{A}{2} + \sin\left(\frac{A}{2} + B\right)\right\}}{2\cos\frac{A}{2}\left\{\cos\frac{A}{2} - \cos\left(\frac{A}{2} + B\right)\right\}}$

$$= \frac{\sin\frac{A}{2}\left\{2\sin\frac{A+B}{2}\cos\frac{B}{2}\right\}}{\cos\frac{A}{2}\left\{2\sin\frac{B}{2}\sin\left(\frac{A+B}{2}\right)\right\}}$$

$$= \tan\frac{A}{2}\cot\frac{B}{2}$$

11. $\dfrac{\cos A}{1 \mp \sin A} = \tan\left(45^\circ \pm \dfrac{A}{2}\right)$

Ans.

Sol. $\dfrac{\cos A}{1 \pm \sin A}$

$$= \frac{\cos^2\frac{A}{2} - \sin^2\frac{A}{2}}{\sin^2\frac{A}{2} + \cos^2\frac{A}{2} \mp 2\sin\frac{A}{2}\cos\frac{A}{2}}$$

$$= \frac{\cos^2\frac{A}{2} - \sin^2\frac{A}{2}}{\left(\cos\frac{A}{2} \mp \sin\frac{A}{2}\right)^2} = \frac{\left(\cos\frac{A}{2} \pm \sin\frac{A}{2}\right)}{\left(\cos\frac{A}{2} \mp \sin\frac{A}{2}\right)}$$ s

$$= \frac{\dfrac{\cos\frac{A}{2} \pm \sin\frac{A}{2}}{\cos\frac{A}{2}}}{\dfrac{\cos\frac{A}{2} \mp \sin\frac{A}{2}}{\cos\frac{A}{2}}} = \frac{1 \pm \tan\frac{A}{2}}{1 \mp \tan\frac{A}{2}}$$

$$= \frac{\tan 45^\circ \pm \tan\frac{A}{2}}{1 \mp \tan 45^\circ \tan\frac{A}{2}}$$

$$= \tan\left(45^\circ \pm \frac{\mathbf{A}}{2}\right)$$

12. $\dfrac{\sec 8A-1}{\sec 4A-1}=\dfrac{\tan 8A}{\tan 2A}$

Ans.

Sol. $\dfrac{\sec 8A-1}{\sec 4A-1}=\dfrac{\dfrac{1}{\cos 8A}-1}{\dfrac{1}{\cos 4A}-1}=\dfrac{\dfrac{1-\cos 8A}{\cos 8A}}{\dfrac{1-\cos 4A}{\cos 4A}}$

$$=\frac{(1-\cos 8A)\cos 4A}{(1-\cos 4A)\cos 8A}=\frac{\left(1-1+2\sin^2 4A\right)\cos 4A}{\left(1-1+2\sin^2 2A\right)\cos 8A}$$

$$=\frac{(2\sin 4A\cos 4A)\sin 4A}{2\sin^2 2A\cos 8A}=\frac{\sin 8A\sin 4A}{\cos 8A\cdot 2\sin^2 2A}$$

$$=\tan 8A\frac{2\sin 2A\cos 2A}{2\sin^2 2A}=\tan 8A\frac{\cos 2A}{\sin 2A}$$

$$=\tan 8A\cot 2A=\frac{\tan 8A}{\tan 2A}$$

13. $\dfrac{1+\tan^2(45^o-A)}{1-\tan^2(45^o-A)}=\cos ec\ 2A$

Ans.

Sol. $\dfrac{1+\tan^2\left(45^\circ-A\right)}{1-\tan^2\left(45^\circ-A\right)}=\dfrac{1}{\cos 2\left(45^\circ-A\right)}$ $\left[\because \cos 2\theta=\dfrac{1-\tan^2\theta}{1+\tan^2\theta}\right]$

$$=\frac{1}{\cos\left(90^\circ-2A\right)}$$

$$=\operatorname{cosec} 2A$$

14. $\dfrac{\sin\alpha+\sin\beta}{\sin\alpha-\sin\beta}=\dfrac{\tan\dfrac{\alpha+\beta}{2}}{\tan\dfrac{\alpha-\beta}{2}}$

Ans.

Sol. $\dfrac{\sin\alpha+\sin\beta}{\sin\alpha-\sin\beta}=\dfrac{2\sin\dfrac{\alpha+\beta}{2}\cos\dfrac{\alpha-\beta}{2}}{2\cos\dfrac{\alpha+\beta}{2}\sin\dfrac{\alpha-\beta}{2}}$

$$=\tan\frac{\alpha+\beta}{2}\cot\frac{\alpha-\beta}{2}=\frac{\tan\dfrac{\alpha+\beta}{2}}{\tan\dfrac{\alpha-\beta}{2}}$$

15. $\dfrac{\sin^2 A-\sin^2 B}{\sin A\cos A-\sin B\cos B}=\tan(A+B)$

Ans.

Sol. $\dfrac{\sin^2 A-\sin^2 B}{\sin A\cdot\cos A-\sin B\cdot\cos B}=\dfrac{2\sin^2 A-2\sin^2 B}{2\sin A\cos A-2\sin B\cos B}$

$$=\frac{1-\cos 2A-(1-\cos 2B)}{\sin 2A-\sin 2B}=\frac{\cos 2B-\cos 2A}{\sin 2A-\sin 2B}$$

$$=\frac{2\sin(A+B)\sin(A-B)}{2\cos(A+B)\sin(A-B)}$$

$$=\tan(A+B)$$

16. $\tan\left(\dfrac{\pi}{4}+\theta\right)-\tan\left(\dfrac{\pi}{4}-\theta\right)=2\tan 2\theta$

Ans.

Sol. $\tan\left(\dfrac{\pi}{4}+\theta\right)-\tan\left(\dfrac{\pi}{4}-\theta\right)=\dfrac{\tan\dfrac{\pi}{4}+\tan\theta}{1-\tan\dfrac{\pi}{4}\tan\theta}-\dfrac{\tan\dfrac{\pi}{4}-\tan\theta}{1+\tan\dfrac{\pi}{4}\tan\theta}$

$$=\frac{1+\tan\theta}{1-\tan\theta}-\frac{1-\tan\theta}{1+\tan\theta}$$

$$=\frac{(1+\tan\theta)^2-(1-\tan\theta)^2}{\left(1-\tan^2\theta\right)}$$

$$=\frac{4\tan\theta}{1-\tan^2\theta}=\frac{2\times 2\tan\theta}{1-\tan^2\theta}$$

$= 2\tan 2\theta$

17. $\dfrac{\cos A+\sin A}{\cos A-\sin A}-\dfrac{\cos A-\sin A}{\cos A+\sin A}=2\tan 2A$

Ans.

Sol. $\dfrac{\cos A+\sin A}{\cos A-\sin A}-\dfrac{\cos A-\sin A}{\cos A+\sin A}$

$=\dfrac{(\cos A+\sin A)^2-(\cos A-\sin A)^2}{\cos^2 A-\sin^2 A}$

$=\dfrac{4\sin A\cos A}{\cos 2A}=\dfrac{2\sin 2A}{\cos 2A}=2\tan 2A$

18. $\cot(A+15^\circ)-\tan(A-15^\circ)=\dfrac{4\cos 2A}{1+2\sin 2A}$

Ans.

Sol. $\cot\left(A+15^\circ\right)-\tan\left(A-15^\circ\right)=\dfrac{\cos\left(A+15^\circ\right)}{\sin\left(A+15^\circ\right)}-\dfrac{\sin\left(A-15^\circ\right)}{\cos\left(A-15^\circ\right)}$

$=\dfrac{\cos\left(A+15^\circ\right)\cos\left(A-15^\circ\right)-\sin\left(A+15^\circ\right)\sin\left(A-15^\circ\right)}{\sin\left(A+15^\circ\right)\cos\left(A-15^\circ\right)}$

$=\dfrac{\cos\left(A+15^\circ+A-15^\circ\right)}{\frac{1}{2}\left[\sin 2A+\sin 30^\circ\right]}=\dfrac{2\cos 2A}{\sin 2A+\frac{1}{2}}$

$=\dfrac{4\cos 2A}{1+2\sin 2A}$

19. $\dfrac{\sin\theta+\sin 2\theta}{1+\cos\theta+\cos 2\theta}=\tan\theta$

Ans.

Sol. $\dfrac{\sin\theta+\sin 2\theta}{1+\cos\theta+\cos 2\theta}=\dfrac{\sin\theta+2\sin\theta\cos\theta}{1+\cos\theta+2\cos^2\theta-1}$

$$= \frac{\sin\theta(1+2\cos\theta)}{\cos\theta(1+2\cos\theta)}$$

$$= \tan\theta$$

20. $\dfrac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta} = \tan\dfrac{\theta}{2}$

Ans.

Sol. $$\frac{1+\sin\theta-\cos\theta}{1+\sin\theta+\cos\theta} = \frac{1+2\sin\frac{\theta}{2}\cos\frac{\theta}{2}-1+2\sin^2\frac{\theta}{2}}{1+2\sin\frac{\theta}{2}\cos\frac{\theta}{2}+2\cos^2\frac{\theta}{2}-1}$$

$$= \frac{2\sin\frac{\theta}{2}\left(\cos\frac{\theta}{2}+\sin\frac{\theta}{2}\right)}{2\cos\frac{\theta}{2}\left(\sin\theta+\cos\frac{\theta}{2}\right)}$$

$$= \tan\frac{\theta}{2}$$

21. $\dfrac{\sin(n+1)A-\sin(n-1)A}{\cos(n+1)A+2\cos nA+\cos(n-1)A} = \tan\dfrac{A}{2}$

Ans.

Sol. L.H.S. $$= \frac{2\cos\frac{(n+1)A+(n-1)A}{2}\sin\frac{(n+1)A-(n-1)A}{2}}{2\cos nA+2\cos\frac{(n+1)A+(n-1)A}{2}\cos\frac{(n+1)A-(n-1)}{2}}$$

$$= \frac{2\cos An\sin A}{2\cos An(1+\cos A)} = \frac{2\sin\frac{A}{2}\cos\frac{A}{2}}{2\cos^2\frac{A}{2}}$$

$$= \tan\frac{A}{2} = \text{R.H.S}$$

22. $\dfrac{\sin(n+1)A+2\sin nA+\sin(n-1)A}{\cos(n-1)A-\cos(n+1)A}=\cot\dfrac{A}{2}$

Ans.

Sol. L.H.S. $=\dfrac{2\sin nA+\left\{2\sin\dfrac{(n+1)A+(n-1)A}{2}\cos\dfrac{(n+1)A-(n-1)A}{2}\right\}}{\left(2\sin\dfrac{(n-1)A+(n+1)A}{2}\sin\dfrac{(n+1)A-(n-1)A}{2}\right)}$

$=\dfrac{2\sin nA(1+\cos A)}{2\sin nA\sin A}=\dfrac{2\cos^2\dfrac{A}{2}}{2\sin\dfrac{A}{2}\cos\dfrac{A}{2}}$

$=\cot\dfrac{A}{2}=\text{R.H.S}$

23. $\sin(2n+1)A\sin A=\sin^2(n+1)A-\sin^2 nA$

Ans.

Sol. $\sin(2n+1)A\sin A=\dfrac{1}{2}[\cos(2nA+A-A)-\cos(2nA+A+A)\}$

$=\dfrac{1}{2}\{\cos 2nA-\cos 2(nA+A)\}$

$=\dfrac{1}{2}\left\{1-2\sin^2 An-1+2\sin^2(A+nA)\right\}$

$=\sin^2(n+1)A-\sin^2 An$

24. $\dfrac{\sin(A+3B)+\sin(3A+B)}{\sin 2A+\sin 2B}=2\cos(A+B)$

Ans.

Sol. L.H.S. $=\dfrac{2\sin(2A+2B)\cos(B-A)}{2\sin(A+B)\cos(A-B)}$

$=\dfrac{2\sin(A+B)\cos(A+B)}{\sin(A+B)}$

$=2\cos(A+B)=\text{R.H.S}$

25. $\sin 3A + \sin 2A - \sin A = 4\sin A \cos \frac{A}{2} \cos \frac{3A}{2}$

Ans.

Sol. $\sin 3A + \sin 2A - \sin A = \sin 2A + \sin 3A - \sin A$

$= 2\sin A \cos A + 2\cos 2A \sin A$

$= 4\sin A \left\{ \cos \frac{3A}{2} \cos \frac{A}{2} \right\}$

$= 4\sin A \cos \frac{3A}{2} \cos \frac{A}{2}$

26. $\tan 2A = (\sec 2A + 1)\sqrt{\sec^2 A - 1}$

Ans.

Sol. $(\sec 2A + 1)\sqrt{\left(\sec^2 A - 1\right)} = \left(\frac{1}{\cos 2A} + 1 \right) \sqrt{\left(1 + \tan^2 A - 1\right)}$

$= \left(\frac{1 + \cos 2A}{\cos 2A} \right) \tan A = \frac{1 + 2\cos^2 A - 1}{\cos 2A} \times \frac{\sin A}{\cos A}$

$= \frac{2\cos^2 A \cdot \sin A}{\cos 2A \cdot \cos A} = \frac{2\cos A \sin A}{\cos 2A}$

$= \tan 2A$

$= \frac{\sin 2A}{\cos 2A}$

27. $\cos^3 2\theta + 3\cos 2\theta = 4(\cos^6 \theta - \sin^6 \theta)$

Ans.

Sol. $\cos^3 2\theta + 3\cos 2\theta = \cos 2\theta \left\{ \cos^2 2\theta + 3 \right\}$

$= \cos 2\theta \{ \cos^2 \theta - \sin^2 \theta)^2 + 3 \}$

$= \cos 2\theta \left\{ \left(\cos^2 \theta + \sin^2 \theta \right)^2 - 4\sin^2 \theta \cos^2 \theta + 3 \right\}$

$= \cos 2\theta \left\{ 1 - 4\sin^2 \theta \cos^2 \theta + 3 \right\}$

$= 4 \left\{ \left(\cos^2 \theta - \sin^2 \theta \right) \left(1 - \sin^2 \theta \cos^2 \theta \right) \right\}$

$$= 4\left[\left(\cos^2\theta - \sin^2\theta\right)\left\{\left(\sin^2\theta + \cos^2\theta\right)^2 - \sin^2\theta\cos^2\theta\right\}\right]$$

$$= 4\left\{\cos^2\theta - \sin^2\theta\right\}\left\{\left(\sin^4\theta + \cos^4\theta + \sin^2\theta\cos^2\theta\right\}\right.$$

$$= 4\left\{\cos^6\theta - \sin^6\theta\right\}$$

28. $1 + \cos^2 2\theta = 2(\cos^4\theta + \sin^4\theta)$

Ans.

Sol. $2\left(\cos^4\theta + \sin^4\theta\right) = \left(\cos^4\theta + \sin^4\theta + 2\cos^2\theta\sin^2\theta\right) + \left(\cos^4\theta + \sin^4\theta - 2\cos^2\theta\sin^2\theta\right)$

$$= \left(\cos^2\theta + \sin^2\theta\right)^2 + \left(\cos^2\theta - \sin^2\theta\right)^2$$

$$= 1 + \cos^2 2\theta$$

29. $\sec^2 A\,(1 + \sec 2A) = 2\sec 2A$

Ans.

Sol. $\sec^2 A(1 + \sec 2A) = \dfrac{1}{\cos^2 A}\left(\dfrac{1 + \cos 2A}{\cos 2A}\right) = \dfrac{1 + 2\cos^2 A - 1}{\cos^2 A\cos 2A}$

$$= \frac{2}{\cos 2A}$$

$$= 2\sec 2A$$

30. $\operatorname{cosec} A - 2\cot 2A\cos A = 2\sin A$

Ans.

Sol. $\operatorname{cosec} A - 2\cot 2A \cdot \cos A = \dfrac{1}{\sin A} - \dfrac{2\cos 2A}{\sin 2A} \times \cos A$

$$= \frac{1}{\sin A} - \frac{2\cos 2A}{2\sin A\cos A} \times \cos A$$

$$= \frac{1}{\sin A} - \frac{\cos 2A}{\sin A} = \frac{1 - \cos 2A}{\sin A}$$

$$= \frac{1 - 1 + 2\sin^2 A}{\sin A}$$

$= 2\sin A$

31. $\cot A = \frac{1}{2}\left(\cot\frac{A}{2} - \tan\frac{A}{2}\right)$

Ans.

Sol. $\cot A = \frac{\cos A}{\sin A} = \frac{\cos^2\frac{A}{2} - \sin^2\frac{A}{2}}{2\sin\frac{A}{2}\cos\frac{A}{2}}$

$= \frac{1}{2}\left[\frac{\cos\frac{A}{2}}{\sin\frac{A}{2}} - \frac{\sin\frac{A}{2}}{\cos\frac{A}{2}}\right]$

$= \frac{1}{2}\left\{\cot\frac{A}{2} - \tan\frac{A}{2}\right\}$

32. $\sin\alpha \sin(60^\circ - \alpha)\sin(60^\circ + \alpha) = \frac{1}{4}\sin 3\alpha$

Ans.

Sol. $\sin\alpha\sin\left(60^\circ - \alpha\right)\sin\left(60^\circ + \alpha\right)$

$= \sin\alpha\left[\frac{1}{2}\left(\cos 2\alpha - \cos 120^\circ\right)\right]$

$= \sin\alpha\left[\frac{1}{2}\left(\cos 2\alpha + \frac{1}{2}\right)\right]$

$= \frac{1}{2}\left\{\sin\alpha\cos 2\alpha + \frac{1}{2}\sin\alpha\right\}$

$= \frac{1}{4}[2\sin\alpha\cos 2\alpha + \sin\alpha]$

$= \frac{1}{4}\{(\sin 3\alpha - \sin\alpha) + \sin\alpha\}$

$= \frac{1}{4}\sin 3\alpha$

33. $\cos\alpha \cos(60^\circ - \alpha) \cos(60^\circ + \alpha) = \frac{1}{4}\cos 3\alpha$

Ans.

Sol. $\cos\alpha \cdot \cos\left(60^\circ + \alpha\right) \cdot \cos\left(60^\circ - \alpha\right)$

$$= \frac{1}{2}\cos\alpha\left\{\cos 120^\circ + \cos 2\alpha\right\}$$

$$= \frac{1}{2}\cos\alpha\left\{\cos 2\alpha - \frac{1}{2}\right\}$$

$$= \frac{1}{4}\cos\alpha\left\{4\cos^2\alpha - 3\right\}$$

$$= \frac{1}{4}\left\{4\cos^3\alpha - 3\cos\alpha\right\}$$

$$= \frac{1}{4}\cos 3\alpha$$

34. $\cot\alpha + \cot(60^\circ + \alpha) - \cot(60^\circ - \alpha) = 3\cot 3\alpha$

Ans.

Sol. $\cot\alpha + \cot\left(60^\circ + \alpha\right) - \cot\left(60^\circ - \alpha\right)$

$$= \cot\alpha + \frac{\cot 60^\circ \cot\alpha - 1}{\cot 60^\circ + \cot\alpha} - \frac{\cot 60^\circ.\cot\alpha + 1}{\cot\alpha - \cot 60^\circ}$$

$$= \cot\alpha + \frac{\frac{1}{\sqrt{3}}\cot\alpha - 1}{\frac{1}{\sqrt{3}} + \cot\alpha} - \frac{\frac{1}{\sqrt{3}}\cot\alpha + 1}{\cot\alpha - \frac{1}{\sqrt{3}}}$$

$$= \cot\alpha + \frac{\cot\alpha - \sqrt{3}}{1 + \sqrt{3}\cot\alpha} - \frac{\cot\alpha + \sqrt{3}}{\sqrt{3}\cot\alpha - 1}$$

$$= \cot\alpha + \frac{-8\cot\alpha}{3\cot^2\alpha - 1}$$

$$= \frac{3\cot^3\alpha - 9\cot\alpha}{3\cot^2\alpha - 1}$$

$$= 3\left[\frac{\cot^3\alpha - 3\cot\alpha}{3\cot^2\alpha - 1}\right] = 3\cot 3\alpha$$

35. $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$

Ans.

Sol. $\cos 20^\circ \cdot \cos 40^\circ \cdot \cos 60^\circ \cdot \cos 80^\circ$

$$= \frac{1}{2}\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{2}.\cos 20^\circ \cos\left(60^\circ - 20^\circ\right)\cos\left(60^\circ + 20^\circ\right)$$

$$= \frac{1}{2}.\frac{1}{4}\cos 60^\circ$$

$$= \frac{1}{16}$$

36. $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ = \frac{3}{16}$

Ans.

Sol. $\sin 20^\circ \cdot \sin 40^\circ \cdot \sin 60^\circ \cdot \sin 80^\circ$

$$= \frac{\sqrt{3}}{2}\sin 20^\circ \sin 40^\circ \sin 80^\circ = \frac{\sqrt{3}}{2}.\sin 20^\circ \sin\left(60^\circ - 20^\circ\right)\sin\left(60^\circ + 20^\circ\right)$$

$$= \frac{\sqrt{3}}{2}.\frac{1}{4}\sin 60^\circ$$

$$= \frac{3}{16}$$

37. $\cos 4\alpha = 1 - 8\cos^2\alpha + 8\cos^4\alpha$

Ans.

Sol. $\cos 4\alpha = 2\cos^2 2\alpha - 1$

$$= 2\left\{\left(2\cos^2\alpha - 1\right)^2\right\} - 1$$

$$= 2\left[4\cos^4\alpha - 4\cos^2\alpha + 1\right] - 1$$

$$= 8\cos^4\alpha - 8\cos^2\alpha + 2\Big] - 1$$

$$= 1 - 8\cos^2\alpha + 8\cos^4\alpha$$

38. $\sin 4\alpha = 4 \sin \alpha \cos^3 \alpha - 4 \cos \alpha \sin^3 \alpha$

Ans.

Sol. $\sin 4\alpha = 2\sin 2\alpha \cos 2\alpha$

$= 2\left\{2\sin\alpha\cos\alpha\left(\cos^2\alpha - \sin^2\alpha\right)\right\}$

$= 4\sin\alpha\cos^3\alpha - 4\cos\alpha\sin^3\alpha$

39. $\cos 6\alpha = 32 \cos^6 \alpha - 48 \cos^4\alpha + 18 \cos^2 \alpha - 1$

Ans.

Sol. $\cos 6\alpha = 2\cos^2 3\alpha - 1$

$= 2\left(4\cos^3\alpha - 3\cos\alpha\right)^2 - 1$

$= 2\left(16\cos^6\alpha + 9\cos^2\alpha - 24\cos^4\alpha\right) - 1$

$= 32\cos^6\alpha + 18\cos^2\alpha - 48\cos^4\alpha - 1$

40. $\tan 3A \tan 2A \tan A = \tan 3A - \tan 2A - \tan A$

Ans.

Sol. We know, $\tan 3A = \tan(2A + A) = \dfrac{\tan 2A + \tan A}{1 - \tan 2A \tan A}$

Cross multiplying, we get

$$\tan 3A - \tan A \tan 2A \tan 3A = \tan 2A + \tan A$$

1. If $\sin\theta = \frac{1}{2}$ and $\sin\phi = \frac{1}{3}$, find the values of $\sin(\theta+\phi)$ and $\sin(2\theta+2\phi)$

यदि $\sin\theta = \frac{1}{2}$ तथा $\sin\phi = \frac{1}{3}$ तो $\sin(\theta+\phi)$ तथा $\sin(2\theta+2\phi)$ का मान ज्ञात करो।

Ans. $\frac{\pm2\sqrt{2}\pm\sqrt{3}}{6}; \frac{\pm7\sqrt{3}\pm4\sqrt{2}}{18}$

Sol. $\cos\theta = \sqrt{(1-\sin^2\theta)} = \sqrt{\left[1-\left(\frac{1}{2}\right)^2\right]} = \pm\frac{\sqrt{3}}{2}$

$$\cos\phi = \sqrt{(1-\sin^2\phi)} = \sqrt{\left[1-\left(\frac{1}{3}\right)\right]^2} = \pm\frac{2\sqrt{2}}{3}$$

$$\sin(\theta+\phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$$

$$= \frac{1}{2}\left(\pm\frac{2\sqrt{2}}{3}\right) \pm \frac{\sqrt{3}}{2}\times\frac{1}{3}$$

$$= \pm\frac{\sqrt{2}}{3} \pm \frac{1}{2\sqrt{3}} = \frac{\pm2\sqrt{2}\pm\sqrt{3}}{6}$$

and $\sin 2(\theta+\phi) = 2\sin(\theta+\phi)\cos(\theta+\phi)$

$$= 2\sin(\theta+\phi)\{\cos\theta\cos\phi - \sin\theta\sin\phi)$$

$$= \frac{\pm2\sqrt{2}+\sqrt{3}}{6}\left(\pm\frac{2\sqrt{2}}{3}\times\frac{\sqrt{3}}{2} - \frac{1}{2}\times\frac{1}{3}\right)$$

$$= \frac{\pm2\sqrt{2}\pm\sqrt{3}}{3}\left(\pm\frac{\sqrt{6}}{3} - \frac{1}{6}\right)$$

$$= \pm\frac{2\sqrt{2}\pm\sqrt{3}}{3}\left(\pm\frac{2\sqrt{6}\pm1}{6}\right)$$

$$= \frac{\pm7\sqrt{3}\pm4\sqrt{2}}{18}$$

2. The tangent of an angle is 2.4. Find its cosecant, the cosecant of half the angle and the cosecant of the supplement of double the angle.

यदि किसी कोण की स्पर्शज्या का मान 2.4 है इसके cosecant अर्द्धकोण के cosecant तथा इसके दुगुने के पूरक कोण d k cosecant का मान ज्ञात करो।

Ans. $\pm\frac{13}{12};\frac{\sqrt{13}}{2}$ or $\pm\frac{\sqrt{13}}{3};\frac{169}{120}$

Sol. Let the given angle be θ

$\therefore \quad \tan\theta = 2.4 = \frac{12}{5}$

$\Rightarrow \cot\theta = \frac{5}{12}$

Hence, $\operatorname{cosec}^2\theta = \cot^2\theta + 1 = \left(\frac{5}{12}\right)^2 + 1 = \frac{169}{144}$

$\therefore \quad \operatorname{cosec}\theta = \pm\frac{13}{12}$

$\Rightarrow \quad \sin\theta = \pm\frac{12}{13}$

$$2\sin\frac{\theta}{2} = \pm\sqrt{(1+\sin\theta)} \pm \sqrt{(1-\sin\theta)}$$

$$= \pm\sqrt{\left(1+\frac{12}{13}\right)} \pm \sqrt{\left(1-\frac{12}{13}\right)}$$

$$= \pm\sqrt{\frac{25}{13}} \pm \sqrt{\frac{1}{13}} = \frac{\pm5\pm1}{\sqrt{13}} = \pm\frac{4}{\sqrt{13}} \text{ or } \pm\frac{6}{\sqrt{13}}$$

$\therefore \quad \sin\frac{\theta}{2} = \pm\frac{2}{\sqrt{13}}$, or $\pm\frac{3}{\sqrt{13}}$

Hence $\operatorname{cosec}\frac{\theta}{2} = \pm\frac{\sqrt{13}}{3}$ or $\pm\frac{\sqrt{13}}{2}$

$$\cos\theta = \sqrt{(1-\sin^2\theta)} = \sqrt{1-\left(\frac{12}{13}\right)^2} = \pm\frac{5}{13}$$

$$\operatorname{cosec}(180^\circ - 2\theta) = \operatorname{cosec} 2\theta$$

$$= \frac{1}{\sin 2\theta} = \frac{1}{2\sin\theta\cos\theta}$$

$$= \frac{1}{2\left(\pm\frac{5}{13}\times\pm\frac{12}{13}\right)} = \frac{169}{120}$$

3. If $\cos\alpha = \frac{11}{61}$ and $\sin\beta = \frac{4}{5}$, find the values of $\sin^2\frac{\alpha-\beta}{2}$ and $\cos^2\frac{\alpha+\beta}{2}$, the angles α and β being positive acute angles.

यदि $\cos\alpha = \frac{11}{61}$ तथा $\sin\beta = \frac{4}{5}$, तो $\sin^2\frac{\alpha-\beta}{2}$ तथा $\cos^2\frac{\alpha+\beta}{2}$, का मान ज्ञात करो जबकि α तथा β धनात्मक न्यून कोण है।

Ans. $\frac{16}{305}; \frac{49}{305}$

Sol. We know, $\sin\alpha = \sqrt{\left(1-\cos^2\alpha\right)} = \sqrt{\left[1-\left(\frac{11}{61}\right)^2\right]} = \frac{60}{61}$

and $\cos\beta = \sqrt{\left(1-\cos^2\beta\right)} = \sqrt{\left[1-\left(\frac{4}{5}\right)^2\right]} = \frac{3}{5}$

Now $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\alpha$

$= \frac{11}{61}\times\frac{3}{5} + \frac{60}{61}\times\frac{4}{5} = \frac{273}{305}$

Now since $2\sin^2\frac{\alpha-\beta}{2} = 1-\cos(\alpha-\beta)$

$= 1-\frac{273}{305} = \frac{32}{305}$

$\therefore$ $\sin^2\frac{\alpha-\beta}{2} = \frac{16}{305};$

Again $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$= \frac{11}{61}\times\frac{3}{5} - \frac{60}{61}\times\frac{4}{5} = -\frac{207}{305}$

$2\cos^2\frac{\alpha+\beta}{2} = 1+\cos(\alpha+\beta)$

$= 1-\frac{207}{305} = \frac{98}{305}$

$\Rightarrow \qquad \cos^2\frac{\alpha+\beta}{2}=\frac{49}{305}$

4. If $\cos\alpha=\frac{3}{5}$ and $\cos\beta=\frac{4}{5}$, find the value of $\cos\frac{\alpha-\beta}{2}$, the angles α and β being positive acute angles.

यदि $\cos\alpha=\frac{3}{5}$ तथा $\cos\beta=\frac{4}{5}$, तो $\cos\frac{\alpha-\beta}{2}$ का मान ज्ञात करो जबकि α तथा β धनात्मक न्यून कोण है।

Ans. $\frac{7}{5\sqrt{2}}$

Sol. $\sin\alpha=\sqrt{\left(1-\cos^2\alpha\right)}=\sqrt{\left(1-\left(\frac{3}{5}\right)^2\right.}=\pm\frac{4}{5}$

$\sin\beta=\sqrt{\left(1-\cos^2\beta\right)}=\sqrt{1-\left(\frac{4}{5}\right)^2}=\pm\frac{3}{5}$

$\cos(\alpha-\beta)=\cos\alpha\cos\beta+\sin\alpha\sin\beta$

$=\frac{3}{5}\times\frac{4}{5}+\frac{3}{5}\times\frac{4}{5}=\frac{24}{25}$

Now, $2\cos^2\left(\frac{\alpha-\beta}{2}\right)=1+\cos(\alpha-\beta)=1+\frac{24}{25}=\frac{49}{25}$

$\therefore \qquad \cos\left(\frac{1-\beta}{2}\right)=\sqrt{\left(\frac{49}{25\times2}\right)}=\frac{7}{5\sqrt{2}}$

5. Given $\sec\theta=1\frac{1}{4}$, find $\tan\frac{\theta}{2}$ and $\tan\theta$.

दिया गया है $\sec\theta=1\frac{1}{4}$, तो $\tan\frac{\theta}{2}$ तथा $\tan\theta$ का मान ज्ञात करो।

Ans. $\pm\frac{1}{3};\pm\frac{3}{4}$

Sol. $\tan^2\theta=\sec^2\theta-1=\left(\frac{5}{4}\right)^2-1=\frac{9}{16}$

$\therefore \quad \tan\theta = \pm\frac{3}{4}$

Again $\cos\theta = \dfrac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}} = \frac{4}{5}$

$\Rightarrow \quad 5 - 5\tan^2\frac{\theta}{2} = 4 + 4\tan^2\frac{\theta}{2}$

$\Rightarrow \quad 9\tan^2\frac{\theta}{2} = 1,$

$\therefore \quad \tan\frac{\theta}{2} = \pm\frac{1}{3}$

6. If $\cos A = 0.28$, find the value of $\tan\frac{A}{2}$

यदि $\cos A = 0.28$ तो $\tan\frac{A}{2}$ का मान ज्ञात करें।

Ans. $\pm\frac{3}{4}$

Sol. $\cos A = \dfrac{1-\tan^2\frac{A}{2}}{1+\tan^2\frac{A}{2}} = .28$,

$\Rightarrow \quad 1.28\tan^2\frac{A}{2} = .72$

$\Rightarrow \quad \tan^2\frac{A}{2} = \frac{72}{128} = \frac{9}{16}$, or $\tan\frac{A}{2} = \pm\frac{3}{4}$

7. Find the values of (1) $\sin 7\frac{1}{2}^\circ$, (2) $\cos 7\frac{1}{2}^\circ$ (3) $\tan 22\frac{1}{2}^\circ$, and (4) $\tan 11\frac{1}{4}^\circ$

निम्न के मान ज्ञात करो (1) $\sin 7\frac{1}{2}^\circ$, (2) $\cos 7\frac{1}{2}^\circ$ (3) $\tan 22\frac{1}{2}^\circ$ तथा (4) $\tan 11\frac{1}{4}^\circ$

Ans. $\frac{\sqrt{4-\sqrt{2}-\sqrt{6}}}{2\sqrt{2}};\frac{\sqrt{4+\sqrt{2}+\sqrt{6}}}{2\sqrt{2}};\sqrt{2}-1;-(\sqrt{2}+1)+\sqrt{4+2\sqrt{2}}$

Sol. (i) $\sin 7\frac{1^\circ}{2}=\sin\frac{15^\circ}{2}=\sqrt{\left(\frac{1-\cos 15^\circ}{2}\right)}$

$$=\sqrt{\left\{\frac{1-\frac{\sqrt{3}+1}{2\sqrt{2}}}{2}\right\}}=\sqrt{\left\{\frac{2\sqrt{2}-\sqrt{3}-1}{4\sqrt{2}}\right\}}=\sqrt{\left(\frac{4-\sqrt{6}-\sqrt{2}}{8}\right)}$$

$$=\frac{\sqrt{4-\sqrt{2}-\sqrt{6}}}{2\sqrt{2}}$$

(ii) $\cos 7\frac{1^\circ}{2}=\cos\frac{15^\circ}{2}=\sqrt{\left(\frac{1+\cos 15^\circ}{2}\right)}$

$$=\sqrt{\left\{\frac{1+\frac{\sqrt{3}+1}{2\sqrt{2}}}{2}\right\}}=\sqrt{\left(\frac{2\sqrt{2}+\sqrt{3}+1}{4\sqrt{2}}\right)}=\frac{\sqrt{\left(4+\sqrt{6}+\sqrt{2}\right)}}{2\sqrt{2}}$$

(iii) We know $\tan 45^\circ=\frac{2\tan\frac{45^\circ}{2}}{1-\tan^2\frac{45^\circ}{2}}=1$

$\Rightarrow$ $\tan^2\frac{45^\circ}{2}+2\tan\frac{45^\circ}{2}-1=0$

$\therefore$ $\tan\frac{45^\circ}{2}=\frac{-2\pm\sqrt{(4+4)}}{2}=-1\pm\sqrt{2}=\sqrt{2}-1$

(iv) $\tan\frac{45^\circ}{2}=\frac{2\tan\frac{45^\circ}{4}}{1-\tan^2\frac{45^\circ}{4}}$

$\Rightarrow\left(\sqrt{2}-1\right)\left(1-\tan^2\frac{45^\circ}{4}\right)=2\tan\frac{45^\circ}{4}$

$\Rightarrow\left(\sqrt{2}-1\right)\tan^2\frac{45^\circ}{4}+2\tan\frac{45^\circ}{4}-\left(\sqrt{2}-1\right)=0$

$\Rightarrow\tan\frac{45^\circ}{4}=\frac{-2\pm\sqrt{4+4\left(\sqrt{2}-1\right)^2}}{2\left(\sqrt{2}-1\right)}$

$=\frac{-2\pm\sqrt{4+4\left(3-2\sqrt{2}\right)}}{2\left(\sqrt{2}-1\right)}$

$=\frac{-2\pm\sqrt{16-8\sqrt{2}}}{2\left(\sqrt{2}-1\right)}$

$=\frac{-1\pm\sqrt{4-2\sqrt{2}}}{\left(\sqrt{2}-1\right)}$

$\therefore\tan\frac{45^\circ}{4}$ cannot be negative.

$\Rightarrow\tan\frac{45^\circ}{4}=\frac{-1+\sqrt{4-2\sqrt{2}}}{\left(\sqrt{2}-1\right)}=\frac{\left(-1+\sqrt{4-2\sqrt{2}}\right)\left(\sqrt{2}+1\right)}{\left(\sqrt{2}-1\right)\left(\sqrt{2}+1\right)}$

$\Rightarrow\tan\frac{45^\circ}{4}=-\left(\sqrt{2}+1\right)+\sqrt{4+2\sqrt{2}}$

8. If $\sin\theta+\sin\phi=a$ and $\cos\theta+\cos\phi=b$, find the value of $\tan\frac{\theta-\phi}{2}$.

यदि $\sin\theta+\sin\phi=a$ तथा $\cos\theta+\cos\phi=b$, तो $\tan\frac{\theta-\phi}{2}$ का मान ज्ञात करो।

Ans. $\sqrt{\frac{4-a^2-b^2}{a^2+b^2}}$

Sol. $\sin\theta + \sin\phi = a$ _____(i)

$\Rightarrow$ $\cos\theta + \cos\phi = b$ _____(ii)

from equation (i), $2\sin\frac{\theta+\phi}{2}\cos\frac{\theta-\phi}{2} = a$ _____(iii)

from equation (ii), $2\cos\frac{\theta+\phi}{2}\cos\frac{\theta-\phi}{2} = b$ _____(iv)

Squaring eqautions (iii) and (iv) and addding, we have

$$4\cos^2\frac{\theta-\phi}{2} = a^2 + b^2$$

$\therefore$ $$\cos^2\frac{\theta-\phi}{2} = \frac{a^2+b^2}{2}$$ _____(v)

From equation (v) we have, $\sin^2\frac{\theta-\phi}{2} = \frac{4-a^2-b^2}{4}$ _____(vi)

Dividing equation (vi) by equation (v), we get

$$\tan^2\frac{\theta-\phi}{2} = \frac{4-a^2-b^2}{a^2+b^2}$$

$\Rightarrow$ $$\tan\frac{\theta-\phi}{2} = \sqrt{\left(\frac{4-a^2-b^2}{a^2+b^2}\right)}$$

Prove that

सिद्ध करो :

9. $(\cos\alpha + \cos\beta)^2 + (\sin\alpha - \sin\beta)^2 = 4\cos^2\frac{\alpha+\beta}{2}$

Ans.

Sol. $(\cos\alpha + \cos\beta)^2 + (\sin\alpha - \sin\beta)^2$

$$= \left(2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}\right)^2 + \left(2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}\right)^2$$

$$= 4\cos^2\frac{\alpha+\beta}{2}\cos^2\frac{\alpha-\beta}{2} + 4\cos^2\frac{\alpha+\beta}{2}\sin^2\frac{\alpha-\beta}{2}$$

$$= 4\cos^2\frac{\alpha+\beta}{2}\left\{\cos^2\frac{\alpha-\beta}{2} + \sin^2\frac{\alpha-\beta}{2}\right\}$$

$$= 4\cos^2\frac{\alpha+\beta}{2}(1)$$

$= 4\cos^2\dfrac{\alpha-\beta}{2}$

10. $(\cos\alpha + \cos\beta)^2 + (\sin\alpha + \sin\beta)^2 = 4\cos^2\dfrac{\alpha-\beta}{2}$

Ans.

Sol. $(\cos\alpha+\cos\beta)^2 + (\sin\alpha+\sin\beta)^2$

$= \cos^2\alpha+\cos^2\beta+2\cos\alpha\cos\beta+\sin^2\alpha+\sin^2\beta+2\sin\alpha\sin\beta$

$= 2 + 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = 2[1+\cos(\alpha-\beta)]$

$= 2\left[1+2\cos^2\dfrac{\alpha-\beta}{2}-1\right]$

$= 4\cos^2\dfrac{\alpha-\beta}{2}$

11. $(\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2 = 4\sin^2\dfrac{\alpha-\beta}{2}$

Ans.

Sol. $(\cos\alpha-\cos\beta)^2 + (\sin\alpha-\sin\beta)^2$

$= 4\sin^2\left(\dfrac{\alpha+\beta}{2}\right)\sin^2\left(\dfrac{\alpha-\beta}{2}\right) + 4\cos^2\left(\dfrac{\alpha+\beta}{2}\right)\sin^2\left(\dfrac{\alpha-\beta}{2}\right)$

$= 4\sin^2\dfrac{\alpha-\beta}{2}\left\{\sin^2\dfrac{\alpha+\beta}{2}+\cos^2\dfrac{\alpha+\beta}{2}\right\}$

$= 4\sin^2\dfrac{\alpha-\beta}{2}$

12. $\sin A = \dfrac{2\tan\dfrac{A}{2}}{1+\tan^2\dfrac{A}{2}}$

Ans.

Sol. $$\frac{2\tan\frac{A}{2}}{1+\tan^2\frac{A}{2}}=\frac{\frac{2\sin A/2}{\cos A/2}}{1+\frac{\sin^2 A/2}{\cos^2 A/2}}=\frac{\frac{2\sin A/2}{\cos A/2}}{\frac{\cos^2 A/2+\sin^2 A/2}{\cos^2 A/2}}$$

$$=2\sin\frac{A}{2}\cos\frac{A}{2}=\sin A$$

13. $$\cos A=\frac{1-\tan^2\frac{A}{2}}{1+\tan^2\frac{A}{2}}$$

Ans.

Sol. $$\frac{1-\tan^2\frac{A}{2}}{1+\tan^2\frac{A}{2}}=\frac{1-\frac{\sin^2 A/2}{\cos^2 A/2}}{1+\frac{\sin^2 A/2}{\cos^2 A/2}}=\frac{\cos^2 A/2-\sin^2 A/2}{\cos^2 A/2+\sin^2 A/2}$$

$$=\cos^2\frac{A}{2}-\sin^2\frac{A}{2}$$

$$=\cos A$$

14. $$\sec\left(\frac{\pi}{4}+\theta\right)\sec\left(\frac{\pi}{4}-\theta\right)=2\sec 2\theta$$

Ans.

Sol. $$\sec\left(\frac{\pi}{4}+\theta\right)\sec\left(\frac{\pi}{4}-\theta\right)=\frac{1}{\cos\left(\frac{\pi}{4}+\theta\right)}\cdot\frac{1}{\cos\left(\frac{\pi}{4}-\theta\right)}$$

$$=\frac{2}{2\cos\left(\frac{\pi}{4}+\theta\right)\cos\left(\frac{\pi}{4}-\theta\right)}=\frac{2}{\cos\left(\frac{\pi}{2}\right)+\cos(2\theta)}$$

$$=\frac{2}{\cos 2\theta}=2\sec 2\theta$$

15. $\tan\left(45^{o}+\frac{A}{2}\right)=\sqrt{\frac{1+\sin A}{1-\sin A}}=\sec A+\tan A$

Ans.

Sol. $\tan\left(45^{\circ}+\frac{A}{2}\right)$

$$=\frac{\tan 45^{\circ}+\tan\frac{A}{2}}{1-\tan 45^{\circ}\tan\frac{A}{2}}=\frac{1+\tan\frac{A}{2}}{1-\tan\frac{A}{2}}$$

$$=\frac{\cos\frac{A}{2}+\sin\frac{A}{2}}{\cos\frac{A}{2}-\sin\frac{A}{2}}=\sqrt{\left(\frac{\left(\cos\frac{A}{2}+\sin\frac{A}{2}\right)^{2}}{\left(\cos\frac{A}{2}-\sin\frac{A}{2}\right)^{2}}\right)}$$

$$=\sqrt{\left(\frac{\cos^{2}\frac{A}{2}+\sin^{2}\frac{A}{2}+2\sin\frac{A}{2}\cos\frac{A}{2}}{\cos^{2}\frac{A}{2}+\sin^{2}\frac{A}{2}-2\sin\frac{A}{2}\cos\frac{A}{2}}\right)}$$

$$=\sqrt{\left(\frac{1+\sin A}{1-\sin A}\right)}$$

$$=\sqrt{\left(\frac{1+\sin A}{1-\sin A}\right)}\times\sqrt{\left(\frac{1+\sin A}{1+\sin A}\right)}=\frac{1+\sin A}{\sqrt{1-\sin^{2}A}}$$

$$=\frac{1+\sin A}{\cos A}=\frac{1}{\cos A}+\frac{\sin A}{\cos A}$$

$$=\sec A+\tan A$$

16. $\sin^{2}\left(\frac{\pi}{8}+\frac{A}{2}\right)-\sin^{2}\left(\frac{\pi}{8}-\frac{A}{2}\right)=\frac{1}{\sqrt{2}}\sin A$

Ans.

Sol. $\sin^{2}\left(\frac{\pi}{8}+\frac{A}{2}\right)-\sin^{2}\left(\frac{\pi}{8}-\frac{A}{2}\right)$

$$=\frac{1}{2}\left[2\sin^{2}\left(\frac{\pi}{8}+\frac{A}{2}\right)-2\sin^{2}\left(\frac{\pi}{8}-\frac{A}{2}\right)\right]$$

$$= \frac{1}{2}\left[1-\cos\left(\frac{\pi}{4}+A\right)-\left\{1-\cos\left(\frac{\pi}{4}-A\right)\right\}\right]$$

$$= \frac{1}{2}\left[\cos\left(\frac{\pi}{4}-A\right)-\cos\left(\frac{\pi}{4}+A\right)\right]$$

$$= \frac{1}{2}\cdot 2\sin\frac{\pi}{4}\cdot\sin A$$

$$= \frac{1}{\sqrt{2}}\cdot\sin A$$

17. $\cos^2\alpha + \cos^2(\alpha + 120^\circ) + \cos^2(\alpha - 120^\circ) = \frac{3}{2}$

Ans.

Sol. $\cos^2\alpha+\cos^2\left(\alpha+120^\circ\right)+\cos^2\left(\alpha-120^\circ\right)$

$$= \frac{1}{2}\left[1+\cos 2\alpha+1+\cos 2\left(\alpha+120^\circ\right)+1+\cos 2\left(\alpha-120^\circ\right)\right]$$

$$= \frac{1}{2}\{3+\cos 2\alpha+\cos\left(2\alpha+240^\circ\right)+\cos\left(2\alpha-240^\circ\right)]$$

$$= \frac{1}{2}\left[3+\cos 2\alpha+2\cos\alpha\cos 240^\circ\right]$$

$$= \frac{1}{2}\left[3+\cos 2\alpha+2\left(-\frac{1}{2}\right)\cos 2\alpha\right]$$

$$= \frac{1}{2}[3+\cos 2\alpha-\cos 2\alpha] = \frac{3}{2}$$

18. $\cos^4\frac{\pi}{8}+\cos^4\frac{3\pi}{8}+\cos^4\frac{5\pi}{8}+\cos^4\frac{7\pi}{8}=\frac{3}{2}$

Ans.

Sol. $\cos^4\frac{\pi}{8}+\cos^4\frac{3\pi}{8}+\cos^4\frac{5\pi}{8}+\cos^4\frac{7\pi}{8}$

$$= \cos^4\frac{\pi}{8}+\cos^4\left(\frac{\pi}{2}-\frac{\pi}{8}\right)+\cos^4\left(\frac{\pi}{2}+\frac{\pi}{8}\right)+\cos^4\left(\pi-\frac{\pi}{8}\right)$$

$$= \cos^4\frac{\pi}{8} + \sin^4\frac{\pi}{8} + \sin^4\frac{\pi}{8} + \cos^4\frac{\pi}{8}$$

$$= 2\left[\cos^4\frac{\pi}{8} + \sin^4\frac{\pi}{8}\right]$$

$$= 2\left[\left(\cos^2\frac{\pi}{8} + \sin^2\frac{\pi}{8}\right)^2 - 2\cos^2\frac{\pi}{8}\sin^2\frac{\pi}{8}\right]$$

$$= 2\left[1 - \frac{1}{2}\cdot\left(\sin 2\cdot\frac{\pi}{8}\right)^2\right] = 2\left[1 - \frac{1}{2}\cdot\left(\sin\frac{\pi}{4}\right)^2\right]$$

$$= 2\left[1 - \frac{1}{2}\cdot\frac{1}{2}\right] = 2\left[\frac{3}{4}\right] = \frac{3}{2}$$

19. $\sin^4\frac{\pi}{8} + \sin^4\frac{3\pi}{8} + \sin^4\frac{5\pi}{8} + \sin^4\frac{7\pi}{8} = \frac{3}{2}$

Ans.

Sol. $\sin^4\frac{\pi}{8} + \sin^4\frac{3\pi}{8} + \sin^4\frac{5\pi}{8} + \sin^4\frac{7\pi}{8}$

$$= \sin^4\frac{\pi}{8} + \sin^4\left(\frac{\pi}{2} - \frac{\pi}{8}\right) + \sin^4\left(\frac{\pi}{2} + \frac{\pi}{8}\right) + \sin^4\left(\pi + \frac{\pi}{8}\right)$$

$$= \sin^4\frac{\pi}{8} + \cos^4\frac{\pi}{8} + \cos^4\frac{\pi}{8} + \sin^4\frac{\pi}{8}$$

$$= 2\left(\sin^4\frac{\pi}{8} + \cos^4\frac{\pi}{8}\right)$$

For Remaining part see above question.

20. $\cos2\theta\cos2\phi + \sin^2(\theta - \phi) - \sin^2(\theta + \phi) = \cos(2\theta + 2\phi)$

Ans.

Sol. $\cos2\theta\cos2\phi + \sin^2(\theta-\phi) - \sin^2(\theta+\phi)$

$= \cos2\theta\cos2\phi + \sin(\theta - \phi + \theta + \phi)\sin(\theta - \phi - \theta - \phi)$

$= \cos2\theta\cos2\phi + \sin2\theta\sin(-2\phi)$

$= \cos2\theta\cos2\phi - \sin2\theta\sin2\phi$

$= \cos(2\theta + 2\phi)$

21. $(\tan 4A + \tan 2A)(1 - \tan^2 3A \tan^2 A) = 2 \tan 3A \sec^2 A$

Ans.

Sol. $(\tan 4A + \tan 2A)\left(1 - \tan^2 3A \tan^2 A\right)$

$$= \left(\frac{\sin 4A}{\cos 4A} + \frac{\sin 2A}{\cos 2A}\right)(1 - \tan 3A \tan A)(1 + \tan 3A \tan A)$$

$$= \left(\frac{\sin 4A \cos 2A + \cos 4A \sin 2A}{\cos 4A \cos 2A}\right)\left(1 - \frac{\sin 3A}{\cos 3A} \times \frac{\sin A}{\cos A}\right)\left(1 + \frac{\sin 3A \sin A}{\cos 3A \cos A}\right)$$

$$= \left(\frac{\sin 6A}{\cos 4A \cos 2A}\right)\left\{\frac{\cos(3A + A)}{\cos 3A \cos A} \times \frac{\cos(3A - A)}{\cos 3A \cos A}\right\}$$

$$= \frac{\sin 6A \cos 4A \cos 2A}{\cos 4A \cos 2A (\cos 3A \cos A)^2}$$

$$= \frac{2 \sin 3A \cos 3A}{\cos^2 3A \cdot \cos^2 A}$$

$$= 2 \tan 3A \sec^2 A$$

22. $\left(1 + \tan\frac{\alpha}{2} - \sec\frac{\alpha}{2}\right)\left(1 + \tan\frac{\alpha}{2} + \sec\frac{\alpha}{2}\right) = \sin\alpha \sec^2\frac{\alpha}{2}$

Ans.

Sol. $\left\{1 + \tan\frac{\alpha}{2} - \sec\frac{\alpha}{2}\right\}\left\{1 + \tan\frac{\alpha}{2} + \sec\frac{\alpha}{2}\right\}$

$$= \left[\left(1 + \tan\frac{\alpha}{2}\right) - \sec\frac{\alpha}{2}\right]\left[\left(1 + \tan\frac{\alpha}{2}\right) + \sec\frac{\alpha}{2}\right]$$

$$= 1 + \tan^2\frac{\alpha}{2} + 2\tan\frac{\alpha}{2} - \sec^2\frac{\alpha}{2} = 2\tan\frac{\alpha}{2}$$

$$= \frac{2\sin\frac{\alpha}{2}}{\cos\frac{\alpha}{2}} \cdot \frac{\cos\frac{\alpha}{2}}{\cos\frac{\alpha}{2}} = \frac{\sin\alpha}{\cos^2\frac{\alpha}{2}}$$

$= \sin\alpha \sec^2 \frac{\alpha}{2}$

23. If A = 340º, prove that $2\sin\frac{A}{2} = -\sqrt{1+\sin A} + \sqrt{1-\sin A}$ and $2\cos\frac{A}{2} = -\sqrt{1+\sin A} - \sqrt{1-\sin A}$

यदि A = 340º, तो सिद्ध कीजिये $2\sin\frac{A}{2} = -\sqrt{1+\sin A} + \sqrt{1-\sin A}$ और

$2\cos\frac{A}{2} = -\sqrt{1+\sin A} - \sqrt{1-\sin A}$

Ans. If $A = 340^\circ$ or $\frac{A}{2} = 170^\circ$

$\cos 170^\circ = \cos(180^\circ - 10^\circ) = -\cos 10^\circ$

and $\sin 170° = \sin 10°$

cos 170° is –ive and sin 170° is +ve but numerical value of cos 170° is greater than sin 170°

So $\sin\frac{A}{2} + \cos\frac{A}{2}$ is –ve

and $\sin\frac{A}{2} - \cos\frac{A}{2}$ is +ve

$\sin\frac{A}{2} + \cos\frac{A}{2} = \sqrt{(1+\sin A)}$ _____(i)

$\sin\frac{A}{2} - \cos\frac{A}{2} = +\sqrt{(1-\sin A)}$ _____(ii)

Adding equation (i) and (ii), we get

$$2\cos\frac{A}{2} = -\sqrt{(1+\sin A)} + \sqrt{(1-\sin A)}$$

Substraction equations (ii) from equation (i), we have

$$2\cos\frac{A}{2} = -\sqrt{(1+\sin A)} + \sqrt{(1-\sin A)}$$

Sol.

24. Within what respective limits must $\frac{A}{2}$ lie when :

(1) $2\sin\frac{A}{2} = \sqrt{1+\sin A} + \sqrt{1-\sin A}$ (2) $2\sin\frac{A}{2} = -\sqrt{1+\sin A} + \sqrt{1-\sin A}$

(3) $2\sin\frac{A}{2}=+\sqrt{1+\sin A}-\sqrt{1-\sin A}$ (4) $2\cos\frac{A}{2}=+\sqrt{1+\sin A}-\sqrt{1-\sin A}$

$\frac{A}{2}$ के मान की सीमाऐं क्या होंगी जबकि

(1) $2\sin\frac{A}{2}=\sqrt{1+\sin A}+\sqrt{1-\sin A}$ (2) $2\sin\frac{A}{2}=-\sqrt{1+\sin A}+\sqrt{1-\sin A}$

(3) $2\sin\frac{A}{2}=+\sqrt{1+\sin A}-\sqrt{1-\sin A}$ (4) $2\cos\frac{A}{2}=+\sqrt{1+\sin A}-\sqrt{1-\sin A}$

Ans. (1) $2n\pi+\frac{\pi}{4}$ and $2n\pi+\frac{3\pi}{4}$ (2) $2n\pi+\frac{3\pi}{4}$ and $2n\pi+\frac{5\pi}{4}$

(3) $2n\pi-\frac{\pi}{4}$ and $2n\pi+\frac{\pi}{4}$ (4) $2n\pi+\frac{\pi}{4}$ and $2n\pi+\frac{3\pi}{4}$

Sol. (i) Since $2\sin\frac{A}{2}=\sqrt{(1+\sin A)}+\sqrt{(1-\sin A)}$

$\therefore$ $\sin\frac{A}{2}+\cos\frac{A}{2}$ and $\sin\frac{A}{2}-\cos\frac{A}{2}$ both are +ve.

Hence $\frac{A}{2}$ must be lie between 45° and 135° or $\frac{A}{2}$ must be lie between $2n\pi+\frac{\pi}{4}$ and $2n\pi+\frac{3\pi}{4}$.

(ii) $\therefore$ $2\sin\frac{A}{2}=-\sqrt{(1+\sin A)}+\sqrt{(1-\sin A)}$

So, $\sin\frac{A}{2}+\cos\frac{A}{2}$ is –ive

and $\sin\frac{A}{2}-\cos\frac{A}{2}$ is +ve

Hence $\frac{A}{2}$ lies between 135° and 225° or between $2n\pi+\frac{3\pi}{4}$ and $2n\pi+\frac{5\pi}{4}$

(iii) $2\sin\frac{A}{2}=\sqrt{1+\sin A}-\sqrt{1-\sin A}$

Hence $\sin\frac{A}{2}+\cos\frac{A}{2}$ is positive.

and $\sin\frac{A}{2}-\cos\frac{A}{2}$ is negative.

Hence $\frac{A}{2}$ must lie between 315° & 45°.

i.e. $2n\pi+\frac{\pi}{4}$ and $2n\pi+\frac{7\pi}{4}$

(iv) Here, $2\cos\frac{A}{2} = -\sqrt{(1+\sin A)} + \sqrt{(1-\sin A)}$

Hence $\sin\frac{A}{2} + \cos\frac{A}{2}$ is positive

and $\sin\frac{A}{2} - \cos\frac{A}{2}$ is also positive.

Hence $\frac{A}{2}$ must lie within the limits 45° and 135°

i.e., $2n\pi + \frac{\pi}{4}$ and $2n\pi + \frac{3\pi}{4}$

If A + B + C = 180º, prove that

यदि A + B + C = 180º, तो सिद्ध कीजिये

1. $\sin 2A + \sin 2B - \sin 2C = 4\cos A \cos B \sin C$

Ans.

Sol. $\sin 2A + \sin 2B - \sin 2C = 2\sin(A+B)\cos(A-B) - 2\sin C\cos C$

$= 2\sin(180^\circ - C)\cos(A-B) - 2\sin C\cos C$

$= 2\sin C\{\cos(A-B) - \cos c\}$

$= 2\sin C\{\cos(A-B) + \cos(A+B)\}$ $\{\because \cos(\pi-(A+B)) = -\cos(A+B)\}$

$= 2\sin C\{2\cos A\cos B)$

$= 4\cos A\cos B\sin C$

2. $\cos 2A + \cos 2B + \cos 2C = -1 - 4\cos A\cos B\cos C$

Ans.

Sol. $\cos 2A + \cos 2B + \cos 2C = 2\cos(A+B)\cos(A-B) + 2\cos^2 C - 1$

$= 2\cos(180^\circ - C)\cos(A-B) + 2\cos^2 C - 1$

$= -1 - 2\cos C\{\cos(A-B) - \cos C\}$

$= -1 - 2\cos C\{\cos(A-B) + \cos(A+B)\}$ $[\because \cos C = -\cos(A+B)]$

$= -1 - 2\cos C\{2\cos A\cos B\}$

$= -1 - 4\cos A\cos B\cos C$

3. $\cos 2A + \cos 2B - \cos 2C = 1 - 4\sin A\sin B\cos C$

Ans.

Sol. $\cos 2A + \cos 2B - \cos 2C = 2\cos(A+B)\cos(A-B) - 2\cos^2 C + 1$

$= 1 + 2\cos(180^\circ - C)\cos(A-B) - 2\cos^2 C$

$= 1 - 2\cos C\{\cos(A-B) + \cos C\}$

$= 1 - 2\cos C\{\cos(A-B) - \cos(A+B)\}$

$= 1 - 2\cos C(2\sin A\sin B\}$

$= 1 - 4\sin A \sin B \cos C$

4. $\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

Ans.

Sol. $\sin A + \sin B + \sin C = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2} + 2\sin\frac{C}{2}\cos\frac{C}{2}$

$= 2\sin\left(90^\circ - \frac{C}{2}\right)\cos\left(\frac{A-B}{2}\right) + 2\sin\frac{C}{2}\cos\frac{C}{2} \quad \left(\because \frac{A}{2} + \frac{B}{2} = 90^\circ - \frac{C}{2}\right)$

$= 2\cos\frac{C}{2}\left\{\cos\left(\frac{A-B}{2}\right) + \sin\frac{C}{2}\right\}$

$= 2\cos\frac{C}{2}\left\{\cos\frac{A-B}{2} + \cos\frac{A+B}{2}\right\}$

$= 4\cos\frac{C}{2}\left\{\cos\frac{A}{2}\cos\frac{B}{2}\right\}$

5. $\sin A + \sin B - \sin C = 4\sin\frac{A}{2}\sin\frac{B}{2}\cos\frac{C}{2}$

Ans.

Sol. $\sin A + \sin B - \sin C = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2} - 2\sin\frac{C}{2}\cos\frac{C}{2}$

$= 2\sin\left(90^\circ - \frac{C}{2}\right)\cos\frac{A-B}{2} - 2\sin\frac{C}{2}\cos\frac{C}{2}$

$= 2\cos\frac{C}{2}\left\{\cos\frac{A-B}{2} - \sin\frac{C}{2}\right\}$

$= 2\cos\frac{C}{2}\left\{\cos\frac{A-B}{2} - \cos\frac{A+B}{2}\right\}$

$= 2\cos\frac{C}{2}\left\{2\sin\frac{A}{2}\sin\frac{B}{2}\right\}$

$= 4\sin\frac{A}{2}\sin\frac{B}{2}\cos\frac{C}{2}$

6. $\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$

Ans.

Sol. $\cos A + \cos B + \cos C = 2\cos \frac{A+B}{2} \cos \frac{A-B}{2} + \cos C$

$= 2\cos\left(90^\circ - \frac{C}{2}\right)\cos \frac{A-B}{2} + 1 - 2\sin^2 \frac{C}{2}$

$= 2\sin \frac{C}{2} \cos \frac{A-B}{2} + 1 - 2\sin^2 \frac{C}{2}$

$= 1 + 2\sin \frac{C}{2}\left\{\cos \frac{A-B}{2} - \sin \frac{C}{2}\right\}$

$= 1 + 2\sin \frac{C}{2}\left\{\cos \frac{A-B}{2} - \cos \frac{A+B}{2}\right\}$

$= 1 + 2\sin \frac{C}{2}\left\{2\sin \frac{A}{2} \sin \frac{B}{2}\right\}$

$= 1 + 4\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$

7. $\sin^2 A + \sin^2 B - \sin^2 C = 2 \sin A \sin B \cos C$

Ans.

Sol. $\sin^2 A + \sin^2 B - \sin^2 C = \frac{1-\cos 2A}{2} + \frac{1-\cos 2B}{2} - \sin^2 C$

$= \frac{1}{2}\left[2 - 2\cos(A+B)\cos(A-B)\right] - \sin^2 C$

$= 1 - \cos(180^\circ - C)\ \cos(A-B) - \sin^2 C$

$= 1 + \cos C \cos(A-B) - 1 + \cos^2 C$

$= \cos C\ [\cos (A-B) + \cos C]$

$= \cos C[\cos (A-B) - \cos (A+B)]$

$= \cos C[2 \sin A \sin B]$

$= 2 \sin A \sin B \cos C$

8. $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2 \cos A \cos B \cos C$

Ans.

Sol. $\cos^2 A + \cos^2 B + \cos^2 C = \cos^2 A + \cos^2 B + \cos^2 C$

$$= \frac{1+\cos 2A}{2} + \frac{1+\cos 2B}{2} + \cos^2 C$$

$$= \frac{1}{2}[2 + \cos 2A + \cos 2B] + \cos^2 C$$

$$= \frac{1}{2}\Big[2 + 2\cos(A+B)\cos(A-B) + \cos^2 C$$

$$= 1 + \cos\left(180^\circ - C\right)\cos(A-B) + \cos^2 C$$

$$= 1 - \cos C\cos(A-B) + \cos^2 C$$

$$= 1 - \cos C[\cos(A-B) - \cos C]$$

$$= 1 - \cos C[\cos(A-B) + \cos(A+B)]$$

$$= 1 - 2\cos A\cos B\cos C$$

9. $\cos^2 A + \cos^2 B - \cos^2 C = 1 - 2\sin A\sin B\cos C$

Ans.

Sol. $\cos^2 A + \cos^2 B - \cos^2 C = \cos^2 A + 1 - \sin^2 B - \cos^2 C$

$$= \cos^2 A - \sin^2 B + 1 - \cos^2 C$$

$$= 1 + \cos(A+B)\cos(A-B) - \cos^2 C$$

$$= 1 + \cos\left(180^\circ - C\right)\cos(A-B) - \cos^2 C$$

$$= 1 - \cos C\{\cos(A-B) + \cos C\}$$

$$= 1 - \cos C\{\cos(A-B) - \cos(A+B)\}$$

$$= 1 - \cos C\{2\sin A\sin B\}$$

$$= 1 - 2\sin A\sin B\cos C$$

10. $\sin^2\frac{A}{2} + \sin^2\frac{B}{2} + \sin^2\frac{C}{2} = 1 - 2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

Ans.

Sol. $\sin^2\frac{A}{2} + \sin^2\frac{B}{2} + \sin^2\frac{C}{2} = 1 - \cos^2\frac{A}{2} + \sin^2\frac{B}{2} + \sin^2\frac{C}{2}$

$$= 1 - \cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) + \sin^2\frac{C}{2}$$

$$= 1 - \cos\left(90^\circ - \frac{C}{2}\right)\cos\frac{A-B}{2} + \sin^2\frac{C}{2}$$

$$= 1 - \sin\frac{C}{2}\left\{\cos\left(\frac{A-B}{2}\right) - \sin\frac{C}{2}\right\}$$

$$= 1 - \sin\frac{C}{2}\left\{\cos\frac{A-B}{2} - \cos\left(\frac{A+B}{2}\right)\right\}$$

$$= 1 - \sin\frac{C}{2}\left\{2\sin\frac{A}{2}\sin\frac{B}{2}\right\}$$

$$= 1 - 2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

11. $\sin^2\frac{A}{2} + \sin^2\frac{B}{2} - \sin^2\frac{C}{2} = 1 - 2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$

Ans.

Sol. $\sin^2\frac{A}{2} + \sin^2\frac{B}{2} + \sin^2\frac{C}{2} = 1 - \cos^2\frac{A}{2} + \sin^2\frac{B}{2} - \sin^2\frac{C}{2}$

$$= 1 - \cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right) - \sin^2\frac{C}{2}$$

$$= 1 - \sin\frac{C}{2}\cos\left(\frac{A-B}{2}\right) - \sin^2\frac{C}{2}$$

$$= 1 - \sin\frac{C}{2}\left\{\cos\frac{A-B}{2} + \sin\frac{C}{2}\right\}$$

$$= 1 - \sin\frac{C}{2}\left\{\cos\frac{A-B}{2} + \cos\frac{A+B}{2}\right\}$$

$$= 1 - \sin\frac{C}{2}\left\{2\cos\frac{A}{2}\cos\frac{B}{2}\right\}$$

$$= 1 - 2\sin\frac{C}{2}\cos\frac{B}{2}\cos\frac{A}{2}$$

12. $\sin\frac{A}{2} + \sin\frac{B}{2} + \sin\frac{C}{2} - 1 = 4\sin\frac{\pi - A}{4}\sin\frac{\pi - B}{4}\sin\frac{\pi - C}{4}$

Ans.

Sol. $\sin\frac{A}{2} + \sin\frac{B}{2} + \sin\frac{C}{2} - 1 = 2\sin\frac{A+B}{4}\cos\frac{A-B}{4} + \sin\frac{C}{2} - 1$

$$= 2\sin\frac{A+B}{4}\cos\frac{A-B}{4} + \cos\left(\frac{\pi}{2} - \frac{C}{2}\right) - 1$$

$$= 2\sin\frac{A+B}{4}\cos\frac{A-B}{4} + \cos\left(\frac{\pi - C}{2}\right) - 1$$

$$= 2\sin\frac{\pi-C}{4}\cos\frac{A-B}{4} - 2\sin^2\frac{\pi-C}{4}$$

$$= 2\sin\frac{\pi-C}{4}\left\{\cos\frac{A-B}{4} - \sin\frac{\pi-C}{4}\right\}$$

$$= 2\sin\frac{\pi-C}{4}\left\{\cos\frac{A-B}{4} - \cos\left(\frac{\pi}{2} - \frac{A+B}{4}\right)\right\}$$

$$= 2\sin\frac{\pi-C}{4}\left[\cos\frac{A-B}{4} - \cos\frac{2\pi-A-B}{4}\right]$$

$$= 2\sin\frac{\pi-C}{4}\left[2\sin\frac{\pi-A}{4}\sin\frac{\pi-B}{4}\right]$$

$$= 4\sin\left(\frac{\pi-C}{4}\right)\sin\left(\frac{\pi-B}{4}\right)\sin\left(\frac{\pi-A}{4}\right)$$

13. $\dfrac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = 8\sin\dfrac{A}{2}\sin\dfrac{B}{2}\sin\dfrac{C}{2}$

Ans.

Sol. $$\frac{\sin 2A + \sin 2B + \sin 2C}{\sin A + \sin B + \sin C} = \frac{2\sin(A+B)\cos(A-B) + 2\sin C\cos C}{\sin A + \sin B + \sin C}$$

$$= \frac{2\sin C\{\cos(A-B) + \cos C\}}{\sin A + \sin B + \sin C}$$

$$= \frac{2\sin C\{\cos(A-B) - \cos(A+B)\}}{\sin A + \sin B + \sin C}$$

$$= \frac{2\sin C \cdot 2\sin A\sin B}{\sin A + \sin B + \sin C}$$

$$= \frac{32\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}}{4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}}$$

We know that, $\sin A + \sin B + \sin C = 4\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$

$\therefore$ L.H.S. $= 8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

14. $\sin(B + C - A) + \sin(C + A - B) + \sin(A + B - C) = 4\sin A \sin B \sin C$

Ans.

Sol. L.H.S. $= 2\sin C\cos(B - A) + \sin(\pi - 2C)$

$= 2\sin C\cos(B - A) + 2\sin C\cos C$

$= 2\sin C\{\cos(B - A) - \cos(A + B)\}$

$= 4\sin A\sin B\sin C$